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Nutrition

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 1

NUTRITION

Nutrition

People are **heterotrophic** organisms and therefore have to take in food by eating plants or animals to obtain energy for growth and recovery.

Food has to be **digested** to release **nutrients** and provide energy.

Energy is required for all activity and life processes.

The human **diet** comprises all the food and liquids that are taken in.

The study of all aspects involving diet is called **nutrition**.

Food is divided into the following main groups:

FOOD GROUP	Sources	Function	Building units
PROTEINS	Meat, fish, eggs, soya	Growth and repair	Amino acids
CARBOHYDRATES	Starches, e.g. bread and pasta Sugars, e.g. cane sugar, honey	Sources of energy	Glucose
FATS	Butter, oils, fat	Source of energy Protection	Fatty acids and Glycerol
MINERALS, VITAMINS AND WATER	Fruit and vegetables	Protection Health	Vitamins Minerals Water

- Dieticians use a different classification with recommended daily allowances (RDA). They recommend that you eat a particular number of portions (shown in brackets) of dairy products (2), grains (2-4), proteins (2-3) and fruit and vegetables (3-5) each day.

Activity: energy per day:

- Study the values in the columns and answer the questions that follow:

INDIVIDUAL

12 year old boy
educator
miner
12 year old girl
female office worker
pregnant woman

kJ per day

12 000
12 500
15 000
9 500
10 000
11 000

1. Why do you think men and women have different kJ requirements?
2. Why do pregnant and lactating women have such a large kJ requirement?
3. Why does a mineworker have such a large kJ requirement?

4. What does kJ stand for?

5. Explain the reduction in the energy requirements of older women.

YOUR FOOD INTAKE

- Complete the following table dealing with your food intake for the past 24 hours. Indicate the food groups by using the codes: Protein (P), Carbohydrates (C), Fats (F).

Breakfast	Code	Lunch	Code	Supper	Code
.....					
.....					
.....					
.....					
.....					
.....					
.....					

1. How many portions of protein have you eaten?

2. How many portions of fruit and vegetables have you eaten?

3. How much water did you drink?

4. Have you followed a balanced diet? If not, explain how you ought to adjust your diet.

5. Draw a pie chart to show the proportion of **carbohydrates**, **proteins** and **fats** that you have eaten over the 24-hour period.

Activity: DEMONSTRATION: FOOD TESTS

- complete the following after the demonstration:

NUTRIENT	REAGENT	POSITIVE TEST
STARCH	IODINE SOLUTION(Yellow-brown solution)	

FATS AND OILS	GREASE-SPOT TEST	
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Assessment

Learning Outcomes 1 : Scientific Investigation

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

We know this when the learner:

- conducts investigations and collects data;
- evaluates data and communicates findings.

Learning Outcomes 2 : Constructing Science knowledge

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

We know this when the learner:

- interprets information.

Memorandum

ACTIVITY: Energy values and graph:

1. Men are more active.
2. A woman's body has to process extra nutrients and operate more processes.
3. Very active.
4. kilojoules – A joule is the unit of work or energy.
5. They have a slower metabolism – are less active.

ACTIVITY: ANALYSING YOUR FOOD INTAKE

Own observations

CLASS ACTIVITY: Food test demonstration

- Test for starch: brown iodine solution. Becomes purplish-black.
- Grease-spot test: translucent greasy spot on paper.

Digestion

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 2

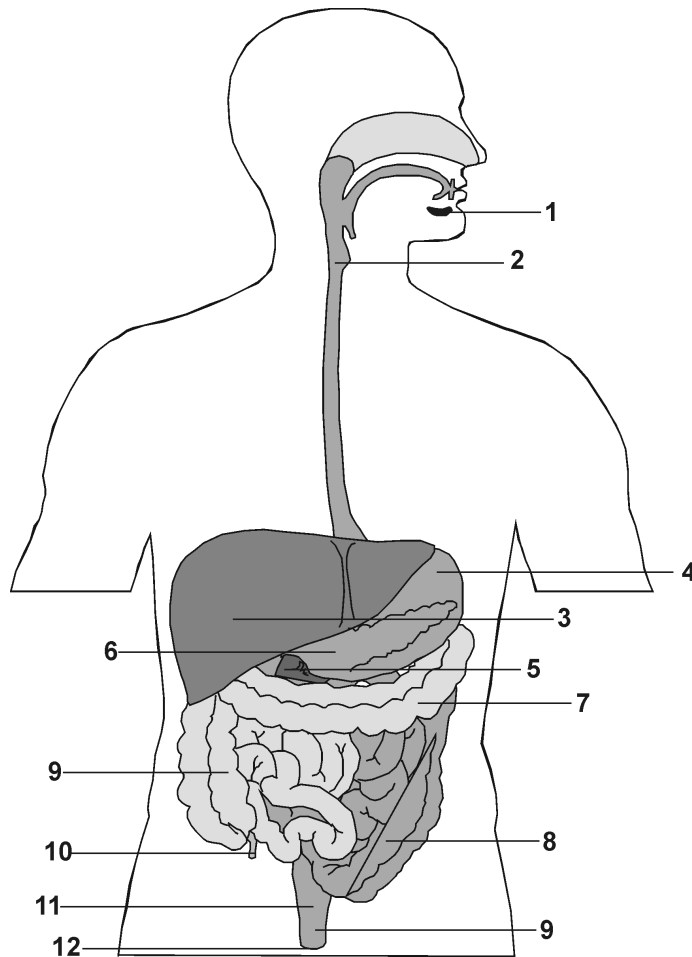
DIGESTION

- Food that is consumed must be broken down into smaller soluble particles that can be absorbed into the blood stream. This process is known as **digestion**.
- The digestive process consists of the following:
 - **Ingestion** – food is taken in
 - **Digestion** – food is broken down
 - **Absorption** – food particles in the bloodstream
 - **Egestion** – excretion of undigested food
- Digestion may be **mechanical** (chewing of teeth and peristalsis) or **chemical** (enzymes).

1. What is peristalsis?
2. Which adaptation of human beings makes peristalsis possible?
3. What are enzymes?
4. Where do enzymes come from?
5. What happens to food particles after they have been absorbed into the bloodstream?

The Human Digestive System

- Study the sketch and write down the captions and their functions:



Regions Of The Digestive Tract

MOUTH:

- 1. Teeth chew food mechanically until fine so that enzymes can operate.
- Saliva from the salivary gland contains mucus to form the chewed food into a **BOLUS**, as well as **AMYLASE**, which digests starch. The

mucus facilitates swallowing.

THROAT TO STOMACH:

- 1. The bolus is swallowed through the pharynx or throat and follows the **oesophagus** or **gullet** to the stomach. **PERISTALSIS** propels the food, even against the force of gravity.
- The **EPIGLOTTIS** prevents food from slipping into the trachea (windpipe) and thus preventing choking.

STOMACH:

- 1. The powerful muscles of the stomach grind the food to **CHYM** within a few hours.
- Enzymes and gastric juice help with the further breakdown of food particles.
- The chym is passed through the **pyloric sphincter** to the small intestine / ileum.

ILEUM:

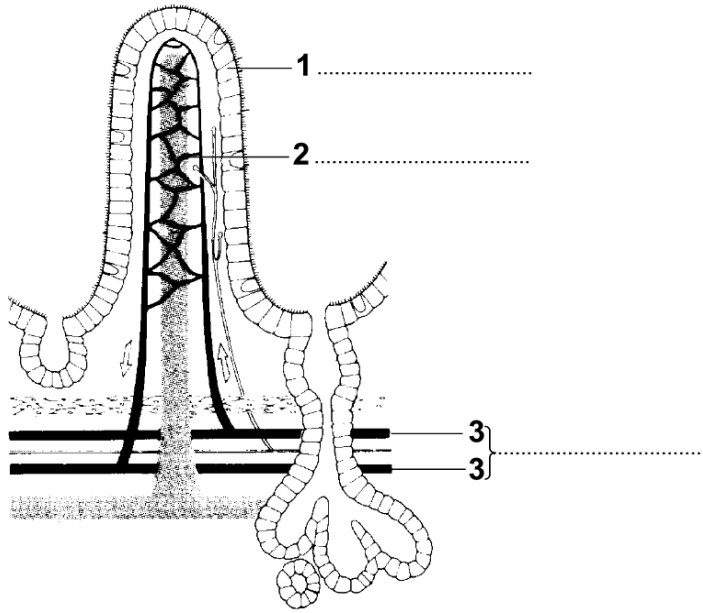
- 1. **GALL** from the liver helps with the digestion of fats by **emulsifying** it.
- **PANCREATIC JUICE** from the pancreas contains many enzymes that assist in completing digestion.
- The intestinal wall is folded and has millions of small projections known as **VILLI**.
- The villi are responsible for the **ABSORPTION** of nutrients.

LARGE INTESTINE:

- 1. This is divided into a rising section, a horizontal section and a descending section.
- The **COLON** reabsorbs water, minerals and vitamins to conserve body fluid.
- Undigested remains are stored temporarily in the last part of the large intestine, the **RECTUM**, until **DEFECATION** occurs through the **ANUS**.
- **ANSWER THE QUESTIONS THAT FOLLOW: DIGESTIVE TRACT**

1. Explain the role of the liver in the processing of toxic substances.
2. Amylose signifies starch. What is amylase?
3. What are the building blocks referred to in an earlier table?
4. What is the gastric juice referred to earlier and from where is it derived?
5. Why doesn't acid destroy the wall of the stomach?
6. What is an ulcer? How does it develop?
7. What is a sphincter?
8. What is a spastic colon and how is it caused?
9. What is the role of roughage in the digestive process?
10. What purpose do the millions of villi in the small intestine serve?
11. Explain how the ileum is adapted to its purpose.
12. Complete the captions to clarify the structure of the villus:

Structure of a villus



Beneficial Bacteria of the Digestive Tract

A variety of harmless bacteria are present in the alimentary canal. Some assist in the digestion of food by producing enzymes, while other bacteria produce vitamin K, which plays a role in blood clotting.

Some bacteria help to prevent infection, while others inhibit the growth of fungi.

All these bacteria exist **SYMBIOTICALLY** in the alimentary canal. The body provides a suitable habitat and they provide the body with the above-mentioned benefits.

Problems occur when this relationship is disturbed, for instance when an **ANTIBIOTIC** is prescribed for a condition, for instance earache.

Remember that antibiotics are effective in the treatment of bacterial infections, but not for viral infections. One should guard against taking more antibiotics than are necessary, and should take them strictly according to the doctor's prescription. It is also important to complete a prescribed course of antibiotics, because ignoring this can assist in the development of so-called "super bugs" that are impervious to all forms of antibiotics.

A simultaneous problem is that antibiotics destroy all forms of bacteria – especially broad-spectrum bacteria, which include the beneficial intestinal bacteria.

Activity: CASE STUDY

- Read the following case study and answer the questions that follow

Victoria is a seven-year-old girl who complained of abdominal pain and lost her appetite. Before these symptoms occurred, she suffered from chronic ear infection for which she had to complete four courses of antibiotics.

She was diagnosed with intestinal dysbiosis (a disturbance in the bacterial population of the alimentary canal). Her diet was adjusted to include lots of yoghurt containing live AB cultures and to cut back on sugar intake. The bacterium *Lactobacillus* is usually added to yoghurt. This is a bacterium that also occurs in the alimentary canal. Victoria was better within days and her ear infection has also cleared up.

1. What caused the disturbance in the alimentary canal?
2. What does “chronic ear infection” mean?
3. Why did Victoria also have to use less sugar?
4. Why did her ear infection improve when her diet was adjusted?
5. Find out what *lacto*- and *bacillus* mean as parts of the name of the bacterium. Why are they printed in italics?

Make your own live yoghurt

1. Heat a litre of fresh milk until bubbles begin to rise – do not allow the milk to come to a boil.
2. Pour it into a thermos flask.
3. Add a tablespoon of fresh yoghurt to the warm milk.

4. Screw down the top of the flask and shake it – leave to stand overnight.
5. Eat with muesli and honey for a healthy breakfast and a daily dose of beneficial bacteria!

Assessment: APPLICATION OF KNOWLEDGE:

Could you apply your knowledge to answer the questions about the case history?

LO 2.4

Assessment

Learning Outcomes 2 : Constructing Science knowledge

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

We know this when the learner:

2.4 applies knowledge.interprets information.

Memorandum

1. contractions of the involuntary muscles in the walls of the digestive tract
2. smooth muscles in the walls of the alimentary canal
3. biological catalysts that accelerate reactions
4. produced in ribosome in cells
5. transported to every cell by the circulatory system

THE HUMAN DIGESTIVE SYSTEM

1. Mucus glands – secretes mucus
2. Oesophagus – connects mouth and stomach peristalsis

3. Liver – detoxifies, stores blood, vitamins, etc.
 4. Stomach – digestion
 5. Gall bladder – stores bile
 6. Pyloric sphincter – allows food to leave the stomach
 7. Pancreas – enzymes and hormones
 8. Duodenum – digestive juices added
 9. Colon – reabsorption of water and minerals
10. Small intestine – absorption of nutrients
11. Appendix – none
12. Rectum – storing food wastes
13. Anus – opening for excretion of waste matter

ANSWER THE FOLLOWING QUESTIONS: ALIMENTARY CANAL

1. The liver detoxifies by removing/absorbing preservatives, colourants, coffee and alcohol.
2. Enzyme that breaks down/catabolises starch.
3. Smallest units into which nutrients can be broken down.
4. Hydrochloric acid – oxyntic cells of the stomach wall.
5. Stomach wall protected by thick mucus membranes.
6. Weak spot in mucus membrane resulting from stress and smoking – enzymes cause the membrane to disintegrate.
7. Muscular valve at lower end of stomach.
8. Muscular spasm of wall of colon – stress.
9. Prevents constipation, stimulates peristalsis.
10. Enlarges the absorbing surface.

11. Long, twisted, villi, microvilli.

12. Labels villum:

CLASS ACTIVITY: CASE STUDY

1. Beneficial bacteria are destroyed by antibiotics.

2. Continued infection.

3. Sugar allows uncontrolled fungi to flourish.

4. Beneficial bacteria help.

5. Lacto – milk; bacillus – rod-shaped.

Nutrition and health

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 3

NUTRITION AND HEALTH

All over the world there are people who are unhealthy because of incorrect nutrition.

Malnutrition can be caused by either too much or too little food, which leads to diseases related to nutrition. These types of malnutrition can be classified as **undernourishment** and **overfeeding**.

Undernourishment:

- 1. Too little nourishment, e.g. **anorexia nervosa** or **bulimia**
- Too few proteins, e.g. **kwashiorkor**
- Too few vitamins, e.g. **scurvy** and **beriberi**

Overfeeding:

- Too much of some kinds of nutriment, e.g. **obesity**

Activity: POSTER

- Collect information on various diseases related to nutrition – divide into groups and make posters to introduce certain illnesses to the class.
- The class must compile and complete a table stating the deficiencies, symptoms and the affected tissues of the following illnesses:
 - Scurvy
 - Beriberi
 - Rachitis/Rickets
 - Pellagra
 - Kwashiokor
 - Anaemia

Discussion on THE HEALTHY DIET

- Discuss different aspects of a healthy diet.
- Name at least five things that should be done to maintain a healthy diet

Assessment: CLASS DISCUSSION AND POSTER

Were you able to supply information and use it to complete the table, poster and question?

LO 1.2; LO 1.3

Assessment

Learning Outcomes 1 : Scientific Investigation

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

We know this when the learner:

- conducts investigations and collects data;
- evaluates data and communicates findings.

Memorandum

CLASS ACTIVITY: POSTER

Name of disease	Deficiency
Scurvy	Vit C
Beriberi	Vit B
Rachitis/rickets	Vit D
Pellagra	Vit B
Kwashiorkor	Proteins
Anaemia	Iron

Tiredness and pallor	

Red blood corpuscles	

DISCUSSION ON THE HEALTHY DIET

1. Eat food from all food groups daily and in correct amounts/proportions.
2. Eat plenty of fruit and vegetables.
3. Drink 8 glasses of water.
4. Avoid refined foods and sugars, as well as coffee and alcohol.
5. Eat small amounts of red meat – remove visible fat.

Human respiratory system

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 4

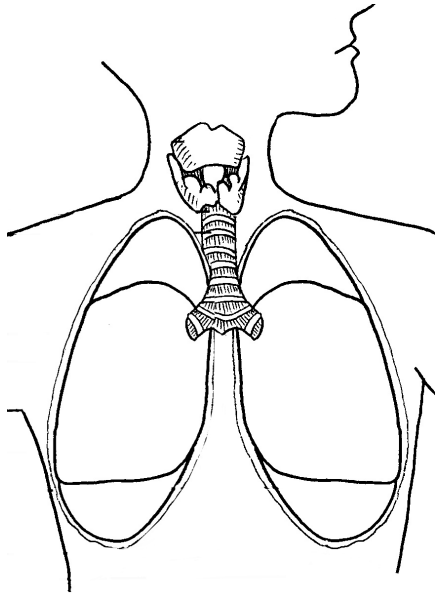
THE HUMAN RESPIRATORY SYSTEM

STRUCTURE OF THE RESPIRATORY SYSTEM:

- You have **air passages** that consist of nasal passages, an oral cavity, throat (pharynx), a **trachea** and two **bronchi**.
- You have **two lungs** consisting of thousands of air sacs called **alveoli**. This increases the area of the lungs to the size of a badminton court.
- The **diaphragm** and **intercostal muscles**.

Class Activity: Study of the lungs of a sheep or a pig

- If possible, your educator should have the lungs of a sheep or a pig in the classroom for you to examine while working through this unit.



the human respiratory system

PROCESSES:

- There are other systems besides the digestive system that are important for the effective functioning of the body. We will examine the systems for respiration, excretion, propulsion and reproduction, as well as the skeleton.
- Every cell in the body of a living organism needs oxygen for **RESPIRATION**. It is the **ENERGY-GIVING PROCESS** during which glucose is “burned” in the presence of oxygen. The waste product of this process is carbon dioxide.
- The part of the cell where this occurs, is the **mitochondria**.
- These substances get to and from the cells by means of the bloodstream.
- **BREATHING** and **EXCHANGE OF GASES** are the processes by which oxygen arrives in the blood and carbon dioxide leaves the blood.
- **BREATHING**: the process by which air moves into and out of the lungs with the help of muscle action. The process of breathing in, or

inhalation, is also known as *INSPIRATION* and breathing out, or exhalation, is known as *EXPIRATION*.

- *EXCHANGE OF GASES*: The process by which gases such as carbon dioxide in the blood and oxygen that is in the air in the lungs are exchanged to get oxygen into the blood and to the cells.
- *RESPIRATION*: the energy-giving process that occurs in the mitochondria in cells. Often people use the term Respiration incorrectly when they mean breathing.

ADAPTATIONS THAT FACILITATE THESE PROCESSES:

- for **BREATHING**, we have a **DIAPHRAGM**, a muscular partition below the lungs that separates the thoracic and abdominal cavities and contracts and relaxes to alter pressure in the thoracic cavity. There are also **INTERCOSTAL MUSCLES** that raise and lower the ribcage.
- for **GASEOUS EXCHANGE** in the lungs, there are thousands of air sacs, or **ALVEOLI**, with very thin, moist walls filled with blood vessels.
- for **GASEOUS EXCHANGE** in the tissues there are fine blood capillaries that extend to each cell, and **tissue fluids** that surround the cells and the mitochondria.
- EXPLAIN EACH OF THE ADAPTATIONS IN THE TABLE:

Diaphragm
Intercostal muscles
Thousands of alveoli

Thin-walled blood capillaries
Alveoli with single-layer epithelia
Mucus membrane with cilia

Class assignment: DRAWING

- Make a sketch to illustrate the alveoli and the blood capillaries that facilitate the movement of gases.

Assessment: DRAWING

Could you draw the sketch correctly?

LO 2.3

Assessment

Learning Outcomes 2 : Constructing Science knowledge

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

We know this when the learner:

- interprets information.

Memorandum

EXPLAIN EACH OF THE ADAPTATIONS LISTED IN THE TABLE:

Increases/decreases the thoracic cavity (chest cavity)
Raise/lower the ribcage
Enlarge the surface
Short distance for diffusion
Thin layer for fast diffusion
Mucus layer traps particles, cilia remove particles and dust

CLASS ASSIGNMENT: DRAWING

- Labels: alveoli

Breathing and health

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 5

Breathing and Health

Air pollution and too much dust and smoke can cause damage to lungs by preventing the cleansing system of the air passages from operating effectively.

When the lung tissue is damaged, oxygen cannot be transferred effectively to the blood, and the heart has to exert itself to supply enough oxygenated blood to all the cells. This leads to health problems and heart disease.

Smoke particles continuously bombard the sensitive lung tissue and this tissue eventually tries to protect itself by growing extra cell layers to act as a shield, which is how **CANCER** develops!

Respiratory Diseases

Emphysema:

- Occurs in people who smoke – the cilia are paralysed and build-up of phlegm occurs
- Continuous coughing begins to damage blood capillaries
- Gas exchange is hampered – the affected person suffers from a lack of oxygen
- Death may result.

Bronchitis:

- Inflammation of the bronchi and bronchioles result from bacterial infections
- Much mucus or phlegm is formed – the sufferer coughs continuously
- Smoking exacerbates the symptoms and recovery is slower.

Asthma:

- An allergic reaction to certain substances
- The sufferer struggles to breathe
- Medication is essential for the dilation of the bronchial tubes
- It is dangerous to use another person's respirator / breathing apparatus. Different respirators have different kinds of chemical substances; therefore it is important to keep to the doctor's prescription.

Tuberculosis:

- Caused by a bacterial infection, which first affects the lungs
- The sufferer coughs continuously, loses weight, loses his or her appetite, has night sweats, is out of breath and coughs up blood.
- It is infectious and bacteria are spread by means of the tiny particles of phlegm that are dispersed during coughing.
- It is completely treatable at the nearest clinic – sufferers merely have to continue the treatment over the extended period for which this might be required.
- It is a problem among inhabitants of the Cape Flats and in poor communities.
- All babies should be vaccinated with the BCG vaccine.

- Questions: LUNG DISORDERS:

1. Why do people with asthma suffer from lack of breath?
2. Why do TB sufferers cough up blood?

3. Why does TB present a problem in poor communities?
4. Why should you not use cough suppressants when you suffer from bronchitis?
5. How does a person suffering from emphysema lose the function of the cilia?

HAVE YOU EVER HAD A PROBLEM WITH CHOKING?

- You can choke when you swallow quickly or when you laugh, talk or cough while swallowing. The epiglottis (a cartilage “lid”), which should close the windpipe temporarily when food has to pass over this opening to the gullet, does not close fast enough and food inadvertently goes down the windpipe or trachea.
- A person who is choking usually tries to get out of the way to hide his discomfort. Follow such a person and find out whether he or she is all right. If they are able to produce a sound, the choking is not too serious and a good fit of coughing will clear the windpipe. However, if the windpipe is blocked, brain damage will occur within 4 to 5 minutes and it is imperative to help such a person.

Doing the HEIMLICH MANOEUVRE!

- Stand behind the sufferer, with your arms around his or her waist.
- Clench one hand into a fist and place the other hand over it, just above the navel and below the ribcage.
- Apply a sharp upward movement repeatedly.
- Small children could be laid face down over the knee.

Demonstrate the Heimlich Manoeuvre to your family, without applying pressure.

You may be able to save a life if you learn to do the Heimlich Manoeuvre correctly. Consider joining your school's First Aid team and following a course in first aid.

Class activity: POSTER on smoking

- Collect warnings from cigarette packets and old magazines – remember that advertising smoking is prohibited!
- Find out about laws concerning smoking – what are the latest developments?
- Create a poster to feature in a consciousness-raising campaign. Put it up outside your classroom, or arrange an exhibition for your school.

DO YOU KNOW THIS?

- Each cigarette shortens the smoker's life by 5,5 minutes – 20 per day therefore causes a loss of six months for every six years of smoking.
- A cigarette contains 4 000 different chemicals. Some of them are deadly toxins, e.g. arsenic, carbon monoxide, nicotine and cyanide!
- Half a gram of nicotine can kill an adult within minutes!

if a smoker stops smoking:

- the capacity for taste and smell will improve within two days
- the morning cough will disappear within two weeks
- the risk of heart disease will be halved within one year

the price shock:

- At almost R13 per packet of 20, a packet of cigarettes per day costs R 390 per month. At this rate, a smoker had to pay R4 680 for his smoking habit in 2004!

Assessment: POSTER

Could you project an effective health warning through sound planning (1.1) and the execution of the investigation (1.2)?

LO 1.1; LO 1.2

Assessment

Learning Outcomes 1 : Scientific Investigation

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

We know this when the learner:

1.1 plans investigations;

- conducts investigations and collects data;

Memorandum

QUESTIONS; PULMONARY DISEASES:

1. Bronchi close up.
2. Blood capillaries rupture because of bouts of coughing.
3. Neglect, unhealthy living conditions, poor education, crowded living conditions.
4. Phlegm must be coughed up.
5. Too much phlegm.

Excretion and the excretory system

NATURAL SCIENCES

Grade 9

EXCRETION AND THE HUMAN EXCRETORY SYSTEM

Grade 6

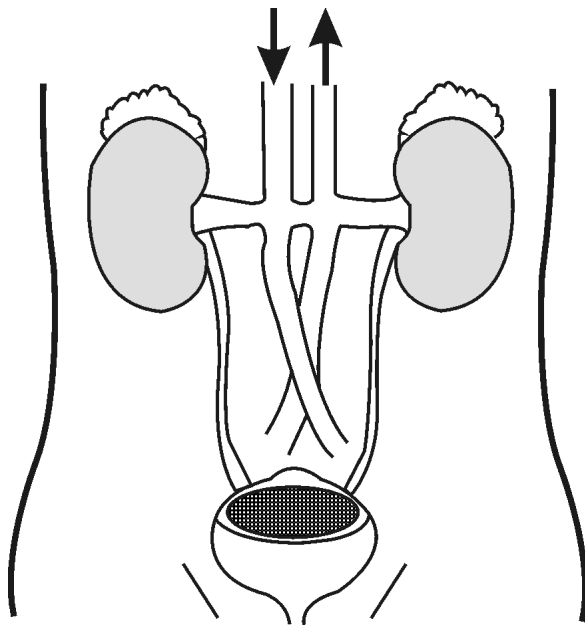
LIFE PROCESSES: HEALTHY LIFE

All waste material that is formed in the cells during metabolism has to be discarded.

The **EXCRETORY SYSTEM** does this. The main excretory system in humans is the **URINARY SYSTEM**, which consists of:

- two kidneys
- two ureters
- the bladder
- a single urethra

URINARY SYSTEM



- **EXCRETION** is the process by which waste is removed.
- **SECRETION** is the process by which beneficial substances like hormones are secreted.
- **EXCRETORY PRODUCTS** are the various substances that have to be cleared away.

Kidneys work like filters and have to filter about 15 000 litres of blood per day. The only filtrate that leaves the body from the kidneys is urine – approximately 1,5 litres per day.

Urine is made up of 96 % water, 2 % urea, 1 % salts and 1 % other metabolic substances.

The kidneys function most efficiently if 6 to 8 glasses of water are consumed during the day. It is also advisable to avoid excessive amounts of coffee, coloured cool drinks and salt. These things make the kidneys work harder.

A person is able to function efficiently on one kidney. People who suffer from acute diseases of the kidneys have to undergo dialysis, which means being linked to a machine (kidney machine) that cleans the blood. Such patients usually are candidates for kidney transplants and are on a waiting list for a donor.

ILLUSTRATION SHOWING THE KIDNEY:

- Draw a simple sketch to illustrate the structure of the kidney. First examine and dissect a pig kidney.

- QUESTIONS DEALING WITH THE KIDNEY:

1. How are the kidneys protected within the body?
2. Explain what kidney stones are and how they are treated nowadays.
3. What is the function of the bladder?
4. Explain the function of the sphincter at the outlet of the bladder.
5. What is incontinence? Have a class discussion about people's sensitivity about this matter.
6. Explain the implications of the following substances in the urine:
 - a) Glucose
 - b) Proteins

Assessment: Sketch of the KIDNEY

Were you able to do the sketch and answer the questions?

LO 2.1

Assessment

Learning Outcomes 2 : Constructing Science knowledge

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

We know this when the learner:

- recalls meaningful information;interprets information.

Memorandum

THE HUMAN EXCRETORY AND SECRETORY SYSTEMS

QUESTIONS ON THE KIDNEYS:

1. layer of fat
2. Uric acid forms crystals – solvent and laser
3. stores urine
4. holds liquid and controls release
5. inability to control the sphincter
6. a) diabetes b) infection the kidneys or pregnancy

Human reproduction

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 7

Human Reproduction

- Reproduction is the process by which offspring are created.
- During this process, **GAMETES** (generative sex cells) have to unite.
- The male gametes are called **SPERM CELLS** and female gametes are **OVA**, or egg cells. Both gametes carry the genetic characteristics of the new individual.

SPERM CELL and OVUM



- In both sexes, gametes only begin to be produced after **PUBERTY**. Hormones from the brain control maturation.
- The cells are formed in the sexual organs, namely in the **TESTES** of the male and in the **OVARIA** of the female.

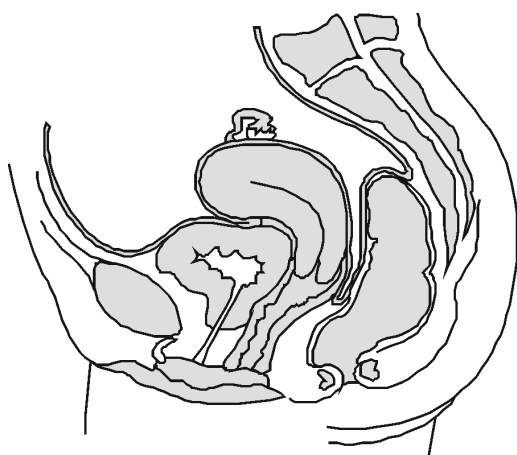
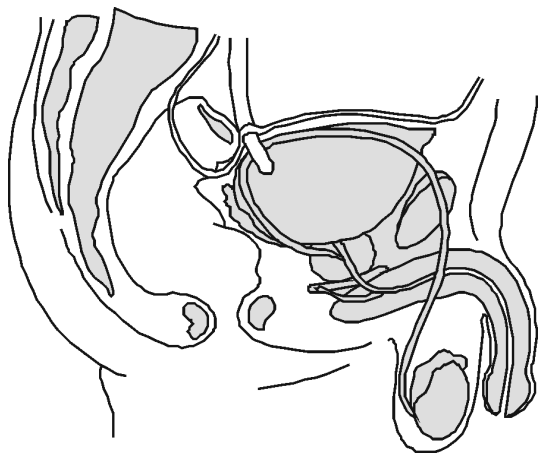
MALE SEXUAL ORGANS:

- 1. 2 **TESTES** in the external **SCROTUM**
- **SPERM TUBES** that carry sperm cells from the testes to the **PROSTATE GLAND** and other glands.
- The **PENIS**, which is responsible for transferring the **SEMEN** (sperm cells and nutrients) to the female system. The release of semen is called **EJACULATION**.

FEMALE SEXUAL ORGANS:

- 1. Two **FALLOPIAN TUBES** for guiding the ova from the ovaries. **FERTILIZATION** normally occurs here.
- A **UTERUS** situated just behind the bladder and in which an infant will develop.
- A **VAGINA** that receives the sperm cells from the penis and which also serves as birth canal.

MALE and FEMALE SEXUAL ORGANS:



The MENSTRUAL CYCLE:

- 1. A cycle that commences in girls once they have reached puberty (commencement varies between the ages of 10 and 15).
- This cycle prepares the female body for possible pregnancy.
- It involves the thickening of the **ENDOMETRIUM** inside the lower portion of the uterus in the area where an embryo will attach itself.
- Various hormones such as **progesterone** and **ESTROGEN** are involved in preparing the body for possible pregnancy. The levels of these hormones rise and recede and cause a variety of changes that

may be very uncomfortable for a girl, e.g. moodiness, tenderness of the breasts, water retention.

- Halfway through the cycle, the egg cell (ovum) is released from the ovary. This is known as **ovulation**.
- The normal duration of the cycle is 28 days. At the end of this period the endometrial layer loosens from the uterus and is discarded. This casting off of the endometrial tissue is known as **MENSTRUATION** and the process takes 3 to 7 days.
- Girls use protective sanitary pads or tampons to absorb the flow of blood.

FERTILIZATION: The COMBINATION of a MALE SPERM CELL with a FEMALE OVUM to form a ZYGOTE

- 1. This occurs in the Fallopian tube
- A special chemical membrane develops immediately to prevent other sperm from penetrating the ovum.

PREGNANCY:

- 1. The zygote immediately begins to divide repeatedly to form an **embryo**.
- A chemical message that travels through the woman's body stimulates further changes.
- The developing **EMBRYO** attaches itself to the **ENDOMETRIUM**, the lining of the wall of the **UTERUS**. This process is known as **IMPLANTATION**.
- A **PLACENTA** develops where the embryo is attached and this nourishes the embryo while it is developing into a foetus. The foetus is attached to the placenta by means of the **UMBILICAL CHORD**.
- The developing foetus is supplied with nutrients and oxygen through the **UMBILICAL CHORD** and waste materials from the foetus, like

- carbon dioxide, are also removed through this chord.
- When the embryo is nine weeks old, it is called a FOETUS.
 - The development of the foetus is completed in approximately 280 days or 9 months.

Class Activity: Discussion on reproduction

1. Explain what is meant by:

a) IMPLANTATION

b) OVULATION

c) FOETUS

d) PUBERTY

e) MENSTRUATION

2. What is the function of the following?

Testis

Ovaries

Penis

Scrotum

Uterus

Fallopian tubes

Placenta

Umbilical chord

3. What is an ECTOPIC PREGNANCY?

4. Explain how the following sets of twins develop:

- a) identical or monozygotic (uniovular) twins;
- b) non-identical or dizygotic (fraternal) twins
- c) Siamese twins

5. What determines whether the baby will be male or female? Explain.

Assessment: CLASS DISCUSSION

Were you able to complete the table and answer the questions?

[LO 1.3]

Assessment

Learning Outcomes 1 : Scientific Investigation

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

We know this when the learner:

- evaluates data and communicates findings.

Memorandum

CLASS ACTIVITY: GROUP DISCUSSION- REPRODUCTION

1. a) implantation of embryo into the wall of the uterus
- b) release of ovum from ovary
- c) developing embryo after two months

d) sexually mature phase in boys and girls

e) shedding of preparatory layer in the monthly cycle

2. Complete:

Umbilical cord	Tube connecting foetus and placenta

3. a pregnancy in which the embryo is implanted in the wall of the Fallopian tube instead of the uterus

4. a) a single zygote that divides early in its development – identical information; always of the same sex

b) two zygotes and two different sperm cells; two different individuals

c) uncompleted identical division

5. father donating X or Y chromosome

Family planning and venereal diseases

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 8

FAMILY PLANNING AND VENEREAL DISEASES

VENEREAL DISEASE:

- As sex involves close contact between two persons, infectious diseases are transferred very easily. **VENEREAL DISEASE** is transferred through **SEXUAL CONTACT**.

Venereal diseases have different symptoms:

AIDS

- The immune system is under attack

The infected person develops illnesses such as cancer, pneumonia and fungal infections and has no defense against them

GONORRHOEA

- Babies are born blind
- Burning sensation when urinating

Painful wrists

SYPHILIS

- Sores on genital organs
- Skin rashes

Bacteria could attack the brain and cause blindness and mental disease

- **QUESTIONS ABOUT VENEREAL DISEASES:**

1. Which of the above diseases can be treated successfully? How?
2. What is the difference between “HIV-positive” and “AIDS”?
3. How could a person who is HIV-positive extend his or her life expectancy?
4. What is NEVIRAPINE and for what is it prescribed?

Assessment: Questions on VENEREAL DISEASES

Were you able to answer the questions successfully?

LO 2.3

FAMILY PLANNING AND CONTRACEPTION:

Couples nowadays have various reasons for wanting to limit the size of their families to fewer or no children and the prevention of pregnancy has become an important issue for them.

Family planning concerns a couple’s planning for pregnancies, usually with the idea of providing financial security for their children.

Many couples, however, simply want to prevent pregnancy altogether and therefore make use of the same **METHODS OF contraception**.

Methods of CONTRACEPTION:

- There are **NATURAL** and **ARTIFICIAL MEANS** of contraception:

Natural:

- **ABSTINENCE** – no sexual intercourse. The popularity of this method by young people is increasing. It is one of a few methods that prevent pregnancy completely (100%) and also prevents sexual diseases.
- **The rhythm method** – During ovulation the body temperature increases. By keeping a record of body temperature, a woman can know when she is ovulating. This information can help couples to avoid, or achieve, pregnancy.

Artificial methods:

CONDOM: a rubber sheath placed over the penis

DIAPHRAGM: a rubber shield that is inserted by a woman before intercourse

IUD (intra-uterine device). A device that is inserted into the uterus by a doctor, and which remains in place for months

CONTRACEPTIVE PILLS: “The Pill”

STERILISATION: Tubes are bound or severed

Class Activity: DISCUSSION ON CONTRACEPTION

1. Why should people use family planning? List three reasons.
2. Explain why the rhythm method is not very dependable as a method of contraception.
3. Why is abstinence gaining popularity as a method of contraception?

4. Explain why none of the artificial methods can be regarded as 100% reliable.
5. Explain what each of the following is:
- a) a VASECTOMY
 - b) a HYSTERECTOMY
 - c) SPERMICIDAL CREAM
6. When you marry, how many children would you like to have, and why?
7. What do you think of ABORTION? Remember that all of us are entitled to our own opinions!

Assessment: CLASS DISCUSSION

Are you able to detect the connection between science and the society in which you live?

SO 3.1

Assessment

Learning Outcomes 2 : Constructing Science Knowledge

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

We know this when the learner:

2.3 interprets information

Learning Outcomes 3: Science, society and environment

The learner will be able to demonstrate an understanding of the interrelationships between science and technology, society and the environment.

We know this when the learner:

3.1 understands science as a human endeavour.

Memorandum

QUESTIONS ABOUT VENEREAL DISEASES:

1. bacterial infections can be treated with antibiotics
2. HIV+ means that a person has been infected with the virus; AIDS is the full-blown disease
3. A healthy lifestyle, exercise, plenty of fresh vegetables and fruit, medication that suppresses the virus infection
4. Prevents mother to child transmission of the virus at and after birth

CLASS ACTIVITY: DISCUSSION ON CONTRACEPTION

1. Quality of life, financial security, costs of living, education
2. cycles may vary – not reliable
3. the only method with real guarantees; return to stronger moral values because of fears related to HIV and other factors
4. All are fallible because of the human aspect.
5. a) severing of sperm tubes of the male – sterilisation
b) removal of the uterus of the female – sterilisation
c) medicinal cream that is usually applied together with other products for contraception
- 6 and 7 own opinion

Growth

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 9

Growth, Development and Adolescence

When an organism **grows**, it shows an increase in mass and size.

An organism **develops** when its cells and tissues take on different forms and functions.

Beginning as a zygote, the growth of a human shows the following steps:

- Morula: a solid ball of cells
- Blastula: a hollow ball of cells beginning to specialise
- Embryo: first two months
- Fetus: from two months

Following birth, the child goes through another series of developmental stages:

- Baby / infant: from birth
- Toddler: from one to two and a half years
- Adolescent: from puberty to adulthood
- Adult: approximately from age 18

Hormones bring about a range of developments:

Boys: TESTOSTERONE

Sexual organs enlarge

Muscles become stronger and larger

Voice deeper – “breaks”

Bodily hair

Girls: ESTROGEN

Sexual organs enlarge

Hips broaden

Hips broaden

Bodily hair

Questions about and Aspects of adolescence: GROUP DISCUSSION

- Some cultures have an initiation process after which the child is regarded as an adult. A range of rituals and various ceremonies are practised. Find out about this and deal sensitively with the knowledge you gain.
- Some cultures and beliefs require circumcision – at differing ages. Here, also, the practices of other people have to be considered sensitively.

1. What is circumcision?

2. What is initiation?

3. What is acne? How is it treated?

4. What is peer pressure? How do you deal with it?

Assessment: GROUP DISCUSSION

Are you able to form a clearer picture of people in their communities?

LO 3.1

Assessment

Learning Outcomes 3: Science, society and environment

The learner will be able to demonstrate an understanding of the interrelationships between science and technology, society and the environment.

We know this when the learner:

3.1 understands science as a human endeavour.

Memorandum

QUESTIONS ABOUT AND ASPECTS OF ADOLESCENCE: GROUP DISCUSSION

1. surgical removal of the foreskin/prepuce of the penis
2. a series of rituals and ceremonies that are practised in some cultures – may include circumcision – male and female
3. skin problems due to abnormal secretion of oils caused by hormones; good basic skin cleansing routine and medicinal creams and preparations

Pressure applied by members of same age group and friends with regard to clothing, behaviour, attitudes and habits; decide for yourself by using sound moral values as a frame of reference; also give consideration to the wisdom and experience of older persons.

Nerves and sensitivity

NATURAL SCIENCES

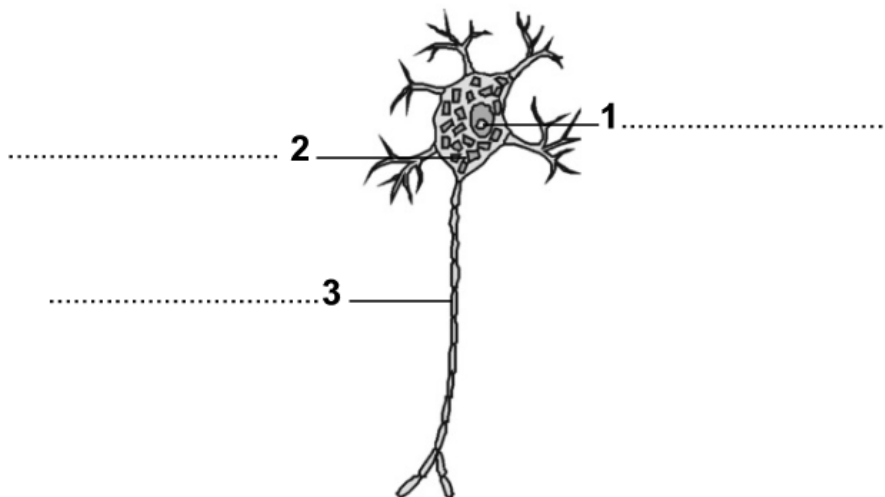
Grade 9

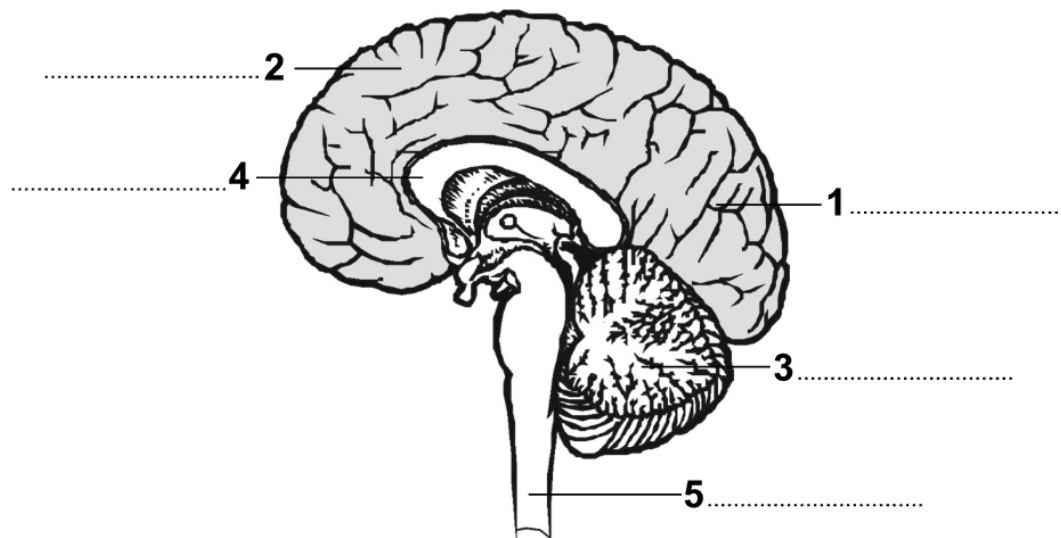
LIFE PROCESSES: HEALTHY LIFE

Module 10

NERVES AND SENSITIVITY

- Sensitivity involves your capacity to react to your changing (internal and external) environment.
1. Indicate the stimulus-reactor path for the process that follows when a bee stings you on the hand.
 2. Add the captions to the illustrations of the brain and the neuron.

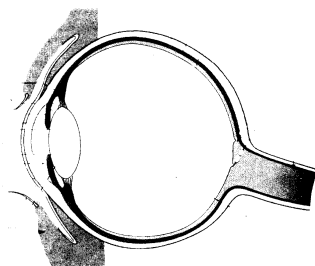




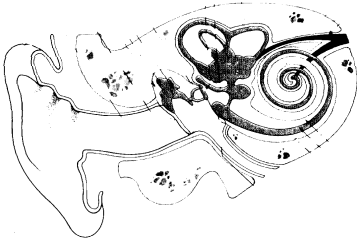
1. Write down the functions of the following:

Brain, spinal cord, motor neuron, sensor neuron

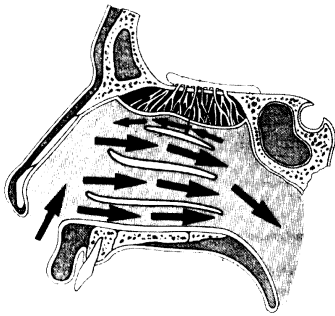
1. Write down the stimulus and sensation of each sense organ:



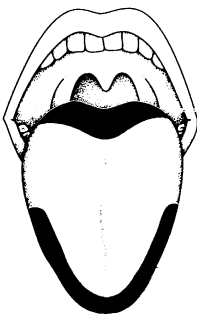
- Light rays penetrate the lens to fall on the retina
- Image is formed
- Cones and rods stimulated
- Impulse transmitted to brain via optic nerve



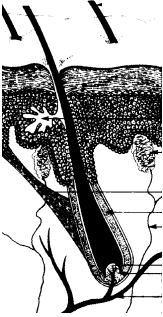
- Sound waves stimulate receptors in cochlea
- Impulse to brain via auditory nerve



- Gas molecule dissolves in mucus membrane
- Olfactory nerve transmits impulse to the brain



- Taste buds on the tongue sense only bitter, sweet, sour or salty (other tastes are perceived by the sense of smell)
- Impulse to brain



- Receptors in skin pick up mechanical, heat, touch, cold or pain stimuli.
- Impulses to brain

5. FIND OUT:

- *Why :*

a) you blink your eyes:

b) your nose runs when you have been crying:

c) your pupils dilate in a dark room:

d) food has no taste when your nose is blocked:

THE EFFECTS OF DRUGS AND ALCOHOL ON THE BRAIN

When your body is in pain, painkillers mask the pain. You can control the nervous system's capacity to feel. This means that medication can be applied usefully, but also that it can be misused.

Drugs are used in this way to provide excitement, but this is dangerous because it can lead to **DEPENDENCY** and result in **TOLERANCE** for particular levels of the drug. When this happens, more and stronger drugs are needed.

PAIN KILLERS:Morphine and heroin

- suppress pain

deaden the senses

STIMULANTS:Caffeine, nicotine, ecstasy anti-depressants

- enhance brain activity
- anxiety (it causes)

lead to depression

TRANQUILLISERS:Alcohol and sleeping tablets

- slow down brain activity
- willpower, judgement and self-control affected

Sluggishness and disturbances of the vision

HALLUCINOGENICS:Dagga and LSD

- feelings of wellbeing
- experiencing things that are not real

Schizophrenia

- Why do young people turn to drugs?

Assessment: APPLICATION

Could you apply your existing knowledge?

LO 2.4

Assessment

Learning Outcomes 2 : Constructing Science knowledge

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

We know this when the learner:

2.4 applies knowledge.

Memorandum

1. impulse/stimulus – receptor – impulse – cns – impulse – effector
2. labels: sketch brain + neurone
4. Complete the table on the senses:
- 5 a) reflex action to lubricate the eye with a lubricant
- b) Tears drain the tear duct that opens at the back of the throat cavity – which is also the back of the nasal cavity
- c) opening enlarges to allow more light
- d) nasal receptors temporarily out of action; tongue only recognises sweetness, saltiness, acidity, and bitterness

Propulsion and transportation

NATURAL SCIENCES

Grade 9

LIFE PROCESSES: HEALTHY LIFE

Module 11

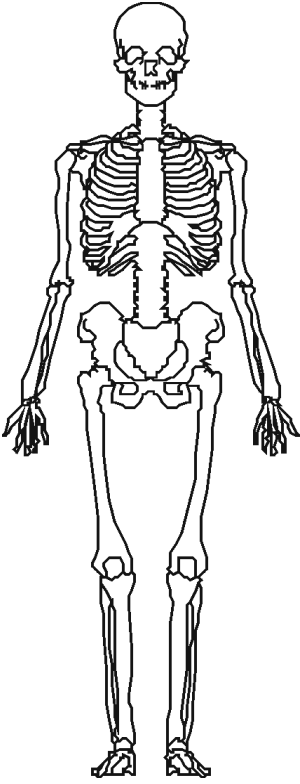
PROPULSION AND TRANSPORTATION

PROPULSION: MUSCLES and SKELETON

People have to be able to move around for the sake of better opportunities.

The propulsion system of humans involves the SKELETON, the JOINTS, and the MUSCLES, LIGAMENTS and TENDONS is a system of lever.

THE SKELETON



Questions: MUSCLES AND SKELETON

1. What happens when you sprain an ankle?
2. Why should ice be applied to a sprained ankle without delay?
3. How would you determine whether someone has broken a leg? What would you do?
4. Explain the difference between a ligament and a tendon.

TRANSPORTATION: THE SYSTEM OF BLOOD VESSELS

All body cells require oxygen and nutrients. We have already studied the role of the respiratory system, so we will be focusing on the **BLOOD**, **BLOOD VESSELS** and the **HEART**.

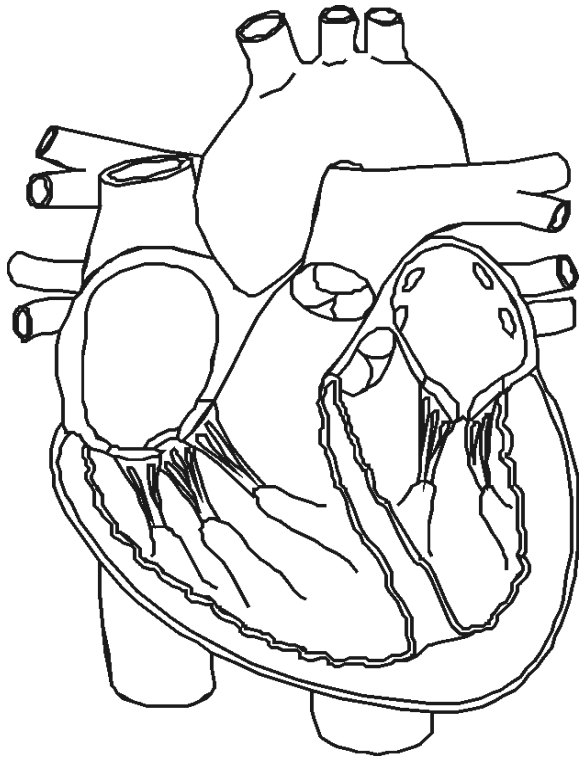
BLOOD VESSELS comprise:

- Capillaries - the delicate, thin-walled blood vessels that surround cells.
- Arteries - transport blood from the heart to the organs and tissues.
- Veins - transport blood from the tissues and organs to the heart.

The HEART:

- A muscular pump that comprises four heart chambers.
- Valves control the flow of blood between the heart chambers.
- The right side of the heart receives the blood from the largest veins, the venae cavae, and directs it to the lungs so that carbon dioxide can be removed and the blood can be charged with oxygen.
- From the lungs, the blood returns to the left side of the heart.
- From here, the blood is pumped through the AORTA to the rest of the body.

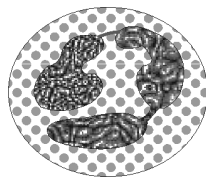
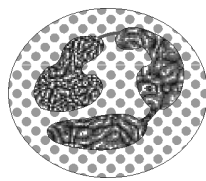
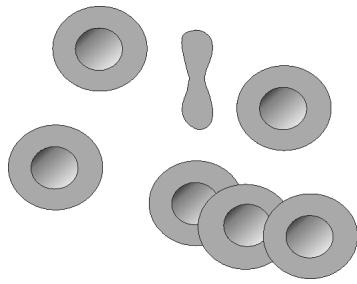
THE HEART



BLOOD:

- Consists of **BLOOD PLASMA** and **CORPUSCLES** (red blood corpuscles, white blood corpuscles and platelets)
- Blood plasma is the medium of transportation – it is a straw-coloured fluid in which everything is in suspension, e.g. gases, waste products, nutrients and hormones.
- Red blood corpuscles are produced in the bone marrow and contain **HAEMOGLOBIN**, the red pigment that transports oxygen in the blood.
- White blood corpuscles attack germs and are also produced in the bone marrow.
- Blood platelets assist blood coagulation and forming scabs.

Red blood corpuscles White blood corpuscles



Questions: THE BLOOD SYSTEM

1. What is leukaemia?
2. Why would a patient who has leukaemia be given bone marrow transplantation?
3. What is anaemia and why do people suffering from this condition receive iron supplements?
4. The walls of arteries have muscles. Why is this so?
5. What is coronary heart disease? Explain the role of cholesterol in this condition.
6. What is the difference between coronary heart disease and a heart attack?
7. Explain the following:

a) Stroke

b) Thrombosis

Assessment: interpretation

Could you apply your existing knowledge to understand other problems?

LO2.3

Assessment

Learning Outcomes 2 : Constructing Science knowledge

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

We know this when the learner:

2.3 interprets information

Memorandum

QUESTIONS: MUSCLES AND SKELETON

1. tendon and ligaments strained – tissue damage
2. limited tissue damage
3. pain; unnatural shape; loss of movement – immobilise; doctor
4. Ligaments attach one bone to another; tendons attach muscles to bones

QUESTIONS: BLOOD SYSTEM / CIRCULATORY SYSTEM

1. leukaemia – white blood corpuscles normally destroy red blood corpuscles

2. This is where blood corpuscles are manufactured
3. anaemia – iron is part of haemoglobin pigment
4. propulsion and pulsation of blood
5. Coronary arteries are blocked because of a coating of cholesterol
6. Coronary heart disease is a result of a condition of the coronary artery and is usually leads to a heart attack which results from an oxygen deficit in the tissues of the cardiac muscle.
7. a) blood vessels on the brain that rupture – brain tissue is damaged as a result of increased pressure
- b) Thrombosis: formation of a blood clot – blocking of the arteries.

Formation of the universe

NATURAL SCIENCES

Grade 9

THE EARTH, GALAXIES AND SPACE PROGRAMMES

Module 12

THE FORMATION OF THE UNIVERSE

At the start of the universe matter did not exist.

According to scientists there was originally only a little ball smaller than an atom of this explosion that became as large as a soccer ball when it started cooling down from infinitely hot to ten billion °C. We don't know what was in the universe before that.

The explanation that the exceedingly hot spaces and a mixture of particles and forces exploded and started expanding, and are still expanding! The theory is that the galaxies are continuing to move apart. Scientists can still detect the microwaves of the afterglow of the explosion.

This is known as the BIG BANG theory. Only one of the four forces in the universe was in operation, namely the super force - it lasted for only a trillionth of a trillionth of a second!

Cosmologists (scientists that study the universe (cosmos)) have a difficult task, as they have nothing against which to test or compare their ideas. That's why the "Big Bang" theory came into being. It is a model that they created to explain all phenomena and aspects of this explosion. According to scientists the universe is still expanding.

A scientist by the name of Edward Hubble recorded this movement of galaxies for the first time in the 1920s. He determined that the speed of a galaxy is directly proportional to its distance from us. This is known as Hubble's Law.

Activity: MAKE A MODEL

- Take an uninflated balloon and paste stars all over its surface. Each star represents a galaxy. Paste a round sticker to represent the Milky Way.
- Now inflate the balloon – note that the galaxies and the Milky Way move further and further apart.

A series of processes in which extreme heat and pressure played a role, led to the formation of matter and elements.

- Hydrogen was formed in the first moments of the formation of the universe after a large number of quarks (Matter: Gr. 8) had been compressed.
 - The universe with all its stars and clouds of matter is comprised almost solely of helium and hydrogen, the smallest and lightest of all elements (as you learnt in Gr. 8).
 - These elements were formed when the quarks blended with the “matter mixture”, resulting in very high pressure and high temperatures.
 - Quarks are so small that 10 billion billion of them next to one another would only stretch one metre.
 - Larger atoms such as beryllium were formed when large numbers of helium atoms were joined together in nuclear reactions within stars.
 - Iron – an important element in the earth's crust – was formed deep inside stars when very high temperatures and pressure compressed carbon atoms. All matter in the universe is held together by FOUR FORCES:
- Gravity
 - Electromagnetic forces
 - Strong nuclear forces
 - Weak nuclear forces

Antimatter also exists – it is like a mirror image of matter and should the two groups come together, they would cancel each other out!!!!
Fortunately there is very little antimatter in space.

When Time Began

- In the beginning, when there was no universe yet, there was also no time. The theory is that it all started as follows:

Time	Events
0 Seconds	• Universe starts as a small, hot ball
<i>10⁻³⁵ seconds</i>	• Space expands at a thousand billion billion billion times a second
<i>10⁻³² seconds</i>	• Space is filled by energy and matter such as quarks
<i>10⁻⁸ seconds</i>	• Matter and antimatter are formed
3 minutes	• Quarks fuse and form hydrogen and helium
1 million	

years	• Gas clouds form long strings
300 million years	• Strings of clouds are formed, which form stars and galaxies

Activity: TEST YOUR KNOWLEDGE

1. What did the universe consist of at the beginning?
2. What is the Big Bang?
3. How did the first elements originate?
4. What is a quark?
5. How did the following elements originate?
 - a) Be
 - b) Fe
6. Name the four most important forces in the universe.
7. What is antimatter and what would happen if it came near matter?

Assessment of understanding of the theory on the origin of the universe:

Do you understand the theory?

[LO 2.3]

Research Assignment 1: TELESCOPES

Write a project of four pages on the use of telescopes and the basic construction of telescopes. Hand your project in on a date set by your educator.

- Find out more, specifically about the:
- Hubble telescope;
- Mcmath solar telescope in Arizona USA;
- giant radio telescope - the Arecibo - In Puerto Rico;
- the SAAO in Sutherland, South Africa;
- the SALT (Southern African Large Telescope) in Sutherland, completed in 2004;
- the radio telescope at Hartbeeshoek, Gauteng.
- Conclude your report with a review of the value of the telescope to science.[LO 1.1; LO 1.2; LO 1.3; LO 3.1]

The Galaxies

- Our sun is only one of an enormous collection of about two billion stars that lie in space in the shape of a fried egg with a diameter of 100 000 **light-years**. If you look up at the sky at night and see a band of small stars, you are looking at the **Milky Way**. This is actually only a view of the edge of our galaxy! The Greek word “galaxy” means “milk”.
- The earth is approximately 300 000 light-years from the centre of our galaxy.
- A **light-year** is one of the most important **UNITS in ASTRONOMY** that you must know. Scientists use such units because *m* and *km* are absurdly small compared to the large distances in space.

Activity: CALCULATIONS

- A light-year = the distance that light travels in one earth-year if it moves at a speed of $300\,000\text{ km/s}^{-1}$

1. Calculate: a light-year

- $365,25\text{ days} \times 24\text{ hours} \times 3\,600\text{ seconds} \times 300\,000\text{ km/s}^{-1} =$
.....
- Therefore: kilometres

- Million, billion or trillion???

NUMBER	10^N	NAME	PREFIX
1 000 000	10^6	MILLION	MEGA
1 000 000 000	10^9	BILLION	GIGA
1 000 000 000 000 000	10^{15}	TRILLION	PETA

2. How many stars are there in our sky?

3. Express 10 billion billion as a number (it will have the prefix **exa**).

4. Another important unit is the **ASTRONOMICAL UNIT(AU)** that represents the distance from the sun to the earth, namely, **150 million kilometres**.

If the diameter of the sun is written in the notation $1,5 \times 10^6$, how would you write the notation of the AU?

5. What is the diameter of our galaxy?

Assessment of calculations:

Were your calculations correct?

[LO 2.4]

Did you know?

The largest formation in the universe is called “the great wall” – a blanket of galaxies that is 500 million light-years long and 16 million light-years

thick.

- Our galaxy is one of billions of such giant star groups or galaxies!!!! This is only what we have been able to establish with our largest and most sensitive telescopes.
- The galaxy nearest to us is the **ANDROMEDA** galaxy. Galaxies comprises groups of stars.
- In between there are giant clouds of gas and particles called **nebulae**. They are the birthplaces of stars.
- The theory is that a hot cloud of gasses - the **solar nebula** - revolved around a newly formed sun four and a half billion years ago and started cooling down. During the cooling-down process billions of drops were formed and started sticking to one another. In this way planets were formed.
- However, it took another half a billion years to cool down sufficiently and form a crust.

Activity : DEFINITIONS

1. A light-year is.
2. The Milky Way is.
3. The Andromeda galaxy is
4. A nebula is
5. The reason why units such as light-years and AU are used is

Assessment of definitions:

Were your definitions correct?

[LO 2.4]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- plans investigations;
- conducts investigations and collects data;
- evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- interprets information;

2.4 applies knowledge.

LO 3: Science, Society and the Environment

The learner will be able to demonstrate an understanding of the interrelationships between science and technology, society and the environment.

This is evident when the learner:

3.1 understands science as a human endeavour;

Memorandum

THE ORIGIN OF THE UNIVERSE

ACTIVITY: QUESTIONS

1. no matter – one little ball smaller than an atom; then a mixture of particles and forces
2. super force by which the smaller ball exploded in matter
3. a number of quarks were compressed to form hydrogen
4. a subatomic particle
5. a) merger of He atoms
b) C atoms under extreme temperatures
6. Force of gravity, electromagnets, powerful nuclear forces and weak nuclear forces
7. opposites of matter cancel out matter

ACTIVITY: CALCULATION

1. 10 million kilometres
2. approximately 2 billion
3. 10^{18}
4. 150×10^6
5. $100\,000 \times 300\,000$

ACTIVITY: DEFINITIONS

1. the distance that light travelling at $300\,000 \text{ km.s}^{-1}$ covers during one earth year
2. an accumulation of billions of stars that include our sun
3. next nearest galaxy

4. gigantic clouds of gas among the galaxies

5. to make it possible to form a concept of a figure

Earth and its features

NATURAL SCIENCE

Grade 9

THE EARTH, GALAXIES AND SPACE PROGRAMMES

Module 13

THE EARTH AND ITS FEATURES

THE EARTH HAS FOUR LAYERS:

- Inner core - very dense, unbelievably hot
- Outer core - in a molten state
- Mantle - molten rock
- Crust - solid rock (very thin)

Activity: DIAGRAM

Draw a circle and draw in these layers . Use different colours to indicate the different layers.

Assessment of diagram:

Did you indicate the layers correctly?

[LO 2.3]

- The core comprises iron and nickel, and the crust consists of oxygen and silicon.
- In the soft mantle there are silicates, oxides and sulphates.

why does the earth rotate?

- The earth rotates because it falls in the direction of the sun. The sun's force of gravity causes the earth to revolve around sun and rotate on its axis.
- It is like a ball rolling down a hill.
- The earth revolves once around its own axis every 24 hours.
- Each planet in the solar system also revolves around the sun in its own orbit.
- The earth takes $365\frac{1}{4}$ days to complete one orbit. This is also called a **revolution** or, in general terms, a year ('**earth-year**').
- It takes each planet a specific number of earth-years to complete one revolution around the sun.
- Mars, has a revolution time of 1,9 earth-years.
- Jupiter takes 11,9 earth-years and Pluto 247,7 earth-years to orbit the sun.

how big is the earth?

split up into groups for the next activity.

Activity: DISCUSSION

information on the earth:

Circumference around poles: 40 008 km

Diameter at poles: 12 714 km

Circumference at Equator: 40 075 km

Diameter at Equator: 12 714 km

1. What can you deduce from these statistics regarding the shape of the earth? What could be the explanation for this?

2. Who was Eratosthenes and what did he say 2 000 years ago about the shape of the earth? Discuss this in groups and write down the answers. (www.union.edu/PUBLIC/GEODEPT/hallocher/sciencelab/Labs/ND2/LN2.htm)

Assessment of DISCUSSION:

[LO 1.2; LO 1.3]

why is life possible on earth?

- It is the only planet in our solar system where there is an atmosphere with oxygen and where there is surface water. Life as we know it requires oxygen and water!
- The oceans take up about 70% of the earth's surface.
- The earth's atmosphere comprises the following gases:
 - oxygen (21%)
 - nitrogen (78%)
 - argon (0.9%)
 - carbon dioxide (0.03%)

Activity: CIRCLE DIAGRAM

Draw a circle. Use different colours, indicating the composition of the atmosphere in percentages in the circle.

Work out the sectors mathematically.

Assessment of circle diagram:

Is your diagram correct? [LO 2.4]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

1.2 conducts investigations and collects data;

- evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- interprets information;

2.4 applies knowledge.

Memorandum

THE EARTH AND ITS FEATURES

ACTIVITY: HOW BIG IS THE EARTH?

1. The shape of the earth is not altogether round. It is slightly flattened at the poles because its rotation is faster at the equator
2. A Greek philosopher who was the person to show that the earth is round by means of a calculation.

ACTIVITY: PIE GRAPH

The atmosphere consists of 21% oxygen, 78% nitrogen and 0.03 % carbon dioxide, and 0.9 % argon

In diagram: **atmospheric gases**

The sun

NATURAL SCIENCES

Grade 9

THE EARTH, GALAXIES AND SPACE PROGRAMMES

Module 14

THE SUN

For us the earth is the centre of our existence and for many years people believed that the earth was also physically the centre of everything. It was Copernicus (1473-1543) who moved away from this **geocentric** view and formulated his **heliocentric** theory. This caused a sensation and forced science in another direction. Johannes Kepler (1571-1630) expanded on this theory and maintained that planets in our solar system revolved in orbits around the sun. He formulated three laws of planetary movement that scientists use today.

The Sun – Our Star

- The sun is a spinning ball of fire (burning hydrogen and helium gasses). Its volume is 1.3 million times that of the earth and it weighs 333 420 times more.
- The pressure in its core is extremely high and converts the sun into a giant nuclear power plant. It splits hydrogen atoms and in this way creates temperatures of up to 15 million °C. This causes the surface of the sun (photosphere) to be turned into an inferno that glows at about 5 500° C.
- The sun has spots that shoot out solar flames of up to approximately 100 000 km. These solar flames can disturb radio waves and therefore interferes with broadcasts.

- We call the atmosphere of the sun the chromosphere. It is surrounded by the corona.
- The SOHO (Solar and Heliospheric Observatory) was launched by NASA in 1995 and now orbits the sun to transmit information about the sun to the earth.

solar eclipses

- SOLAR ECLIPSES occur when the moon comes between the earth and the sun. The eclipse is often only partial and then it is like dusk, as it was on 21 June 2001. During a total eclipse of the sun it becomes completely dark. Such an eclipse occurred in the northern part of our country on 4 December 2002, when it became completely dark.
- Any given town or place on earth will experience about 40 lunar eclipses and 20 partial solar eclipses in any 50-year period. A total eclipse of the sun such as the one in December 2002 occurs only once every 400 years in a particular place.
- The next total solar eclipse in South Africa will only occur in November 2030.
- The dark part of the eclipse is called the **umbra (U)** – or area of total shadow.
- The outer parts are only partly shaded and are called the **penumbra (P)**.
- A total eclipse lasts 7,5 minutes, as the earth rotates and the observer moves out of shadow.
- Only people in the path of the shadow of the eclipse will be able to observe the eclipse.
- When observing a solar eclipse one has to protect ones eyes. Permanent damage to eyes can be caused if the right precautions are not taken.
- Special spectacles can be made or bought. A simple method is to take a piece of cardboard that will fit over your eyes and cut out a small ‘window’. The window should be covered with a double layer of the foil in which teabags are packaged.

Research assignment 2:

Visit the following websites to learn more about solar eclipses:

- <http://sunearth.gsfc.nasa.gov/eclipse/TSE2002/TSE2002.html>
- www.exploratorium.edu/eclipse/

Also find out more about various beliefs regarding solar eclipses in ancient times.

What did the Chinese believe? And the Tahitians?

Pythagoras, the Greek, was the first to explain eclipses. Why do you think this was?

Assessment of Research Assignment:

Did you a) plan how you would obtain the information, b) collect information and

c) communicate information and findings?

[LO 1.1; LO 1.2; LO 1.3]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- plans investigations;
- conducts investigations and collects data;
- evaluates data and communicates findings.

Memorandum

Research

The planets

NATURAL SCIENCES

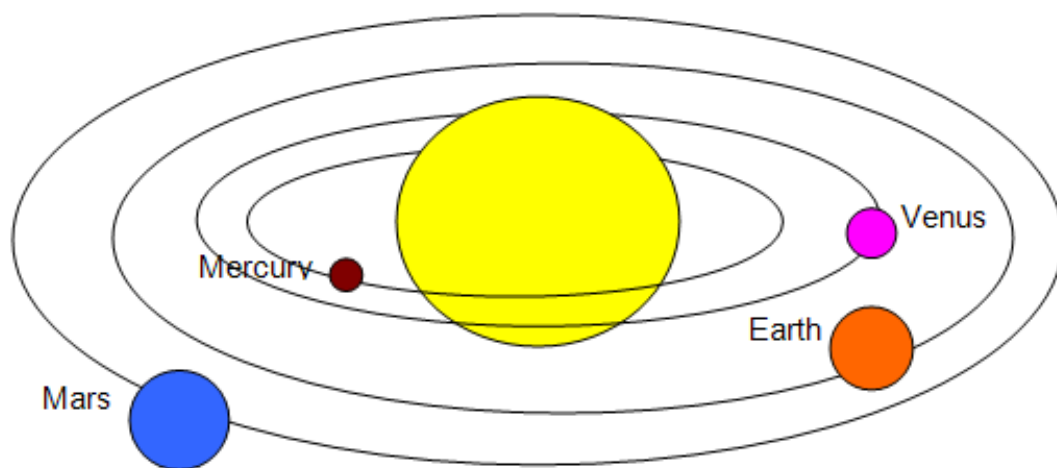
Grade 9

THE EARTH,GALAXIES ANS SPACE PROGRAMMES

Module 15

PLANETS OF OUR SOLAR SYSTEM

The following diagram shows the inner planets of our solar system:



Statistics of

Statistics of the inner planets

Name	Mercury	Venus	Mars	Earth
Diameter (km)	4 878	12 104	6 787	12 750
Mass as % of the earth	0,06	0,8	0,1	-
Distance from sun in millions of km)	58	108	228	150
Number of moons	0	0	2	1
Other facts	▶ No atmosphere	▶ Clouds of sulphuric acid ▶ Thick CO₂ traps heat and causes surface temperature of 480°C ▶ Revolves in opposite direction to all other planets	▶ The red planet (as a result of iron) ▶ Temperature between -125°C and 20°C ▶ Signs of water vapour	

Activity: INTERPRETATION

1. How many planets revolve around the sun?

.....

2. Which planet is closest to the sun?

3. Which planet is furthest from the sun in the diagram?

.....

1. What is the shape of the planets' orbits?

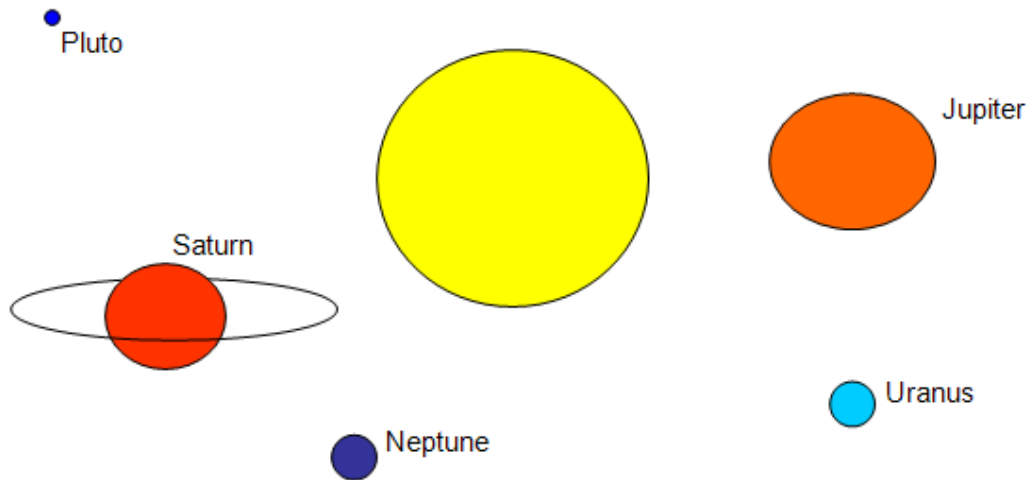
5. Which planet is called the earth's twin because it looks the same from space?

6. Which planet of this group has features that show the greatest resemblance to those of the earth?

Activity: Draw the orbits

[LU 2.1]

The following diagram shows the outer planets of our solar system.



Statistics of the outer planets

<i>Name</i>	Jupiter	Saturn	Uranus	Neptune	Pluto
<i>Diameter (km)</i>	142 800	120 660	51 118	49 528	2 300
<i>Mass times that of the earth</i>	318	95	14.5	17	0,0025
<i>Distance from the sun (in millions of km)</i>	778	1 429	2 871	4 500	5 913
<i>Number of moons</i>	16	18	15	8	1
<i>Other facts</i>	▶ Gas clouds cover the entire planet ▶ Has a large red spot – presumably a large hurricane ▶ Galileo Galilei spotted four of the moons as early as the late 1500s. ▶ The largest moon is bigger than Mercury.	▶ Titan, a moon, has its own atmosphere ▶ Rings of broken pieces of ice surround the planet	▶ An ice giant. The atmosphere is -197°C	▶ Winds blow at 1400 kph ▶ -225°C	▶ -223°C

Activity: QUESTIONS

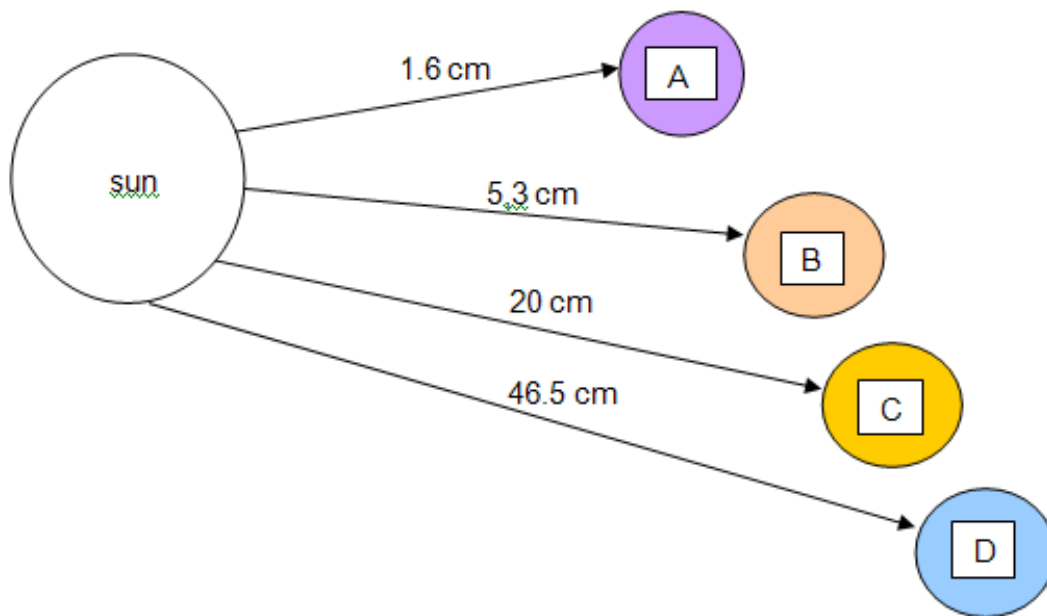
Which planet ...

1. has a red spot?
2. has rings of ice?
3. has the strongest winds?.
4. has the largest mass?
5. has the most moons?
6. is the coldest?

7. has the largest diameter?
8. has the highest temperature?
9. has a moon with an atmosphere? .
10. has a moon larger than another planet?

Assessment of QUESTIONS ON PLANETS: Did you answer the questions correctly? **[LO 2.3]**

Activity: CALCULATING DISTANCES (interpretation task)



Assignment: INTERPRETATION TASK

1. Study the above values and suggested distances from the sun.

1 cm = 1 Astronomical unit

Identify the planets:

A _____

B _____

C _____

D _____

2. Arrange the planets from the largest to the smallest.

3. What planet is 5 times further than the earth from the sun?

4. What planet is 30 times further than the earth from the sun?

5. The gravity of Mars is 0,39 that of the earth. If you weigh 100 kg on earth, you would weigh $100 \times 0,39$ kg, i.e. 39 kg, on Mars.

How much do you weigh on earth?

Calculate how much you would weigh on the following planets and complete the table:

<u>PlanetSurface</u> gravity	Your weight
Mercury0,39	
Venus0,91	
Earth1,00	
Mars0,39	
Jupiter2,60	
Saturn1,10	
Uranus0,88	
Neptune1,14	
Pluto0,05 (approximately)	

6. On which planet would you:

(a) weigh the least?

(b) weigh the most?

Assessment of INTERPRETATION TASK:

Did you do the conversions correctly? Could you categorise the information?

[LO 2.2; LO 2.4]

Activity : GROUP WORK

CONSTRUCT A SOLAR SYSTEM

- You can work in groups to construct
- a complete solar system
- only the inner planets

- only the outer planets
- individual planets
- The project must be three-dimensional, colourful and of use to the educator.
- Suggestion: Use paper maché and water paint.
- Colours must be selected according to the features of the planet.

Assessment of CONSTRUCTION PROJECT:

Did you a) plan how you would obtain information, b) collect information and c) communicate information and findings?

[LO 1.1; LO 1.2; LO 1.3; LO 3.1]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- plans investigations;
- conducts investigations and collects data;
- evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;
- categorises information;

- interprets information;

2.4 applies knowledge.

LO 3: Science, Society and the Environment

The learner will be able to demonstrate an understanding of the interrelationships between science and technology, society and the environment.

This is evident when the learner:

3.1 understands science as a human endeavour.

Memorandum

THE PLANETS OF OUR SOLAR SYSTEM

ACTIVITY: INTERPRETATION

1. 9

2. Mercury

3. Neptune

4. elliptical

5. Venus

6. Mars

ACTIVITY: QUESTIONS – Which planet....

1. Jupiter

2. Saturn

3. Neptune

4. Jupiter

5. Saturn

6. Neptune

7. Jupiter

8. Mercury

9. Saturn's Titan

10. Ganymede, of Jupiter, is the largest one in our galaxy and is bigger than Mercury

ACTIVITY: CALCULATING DISTANCES (translation)

1. A = Mars (1,6 x 150 milj) B= Jupiter C = Uranus D= Pluto

2. Jupiter, Saturn, Uranus, Neptune, Earth, Venus, Mars, Mercury, and Pluto

3. Jupiter

4. Neptune

5. Each according to its own mass

6a) Pluto

6b) Jupiter

Comets

NATURAL SCIENCES

Grade 9

THE EARTH, GALAXIES AND SPACE PROGRAMMES

Module 16

COMETS AND OTHER TRAVELLERS IN SPACE

Comets are pieces of ice and rock that travel in orbits around the sun. According to scientists they are remains of matter from the creation of our solar system. However, most of the remains nearest to the sun have been destroyed. Only those in the cold parts behind Neptune and Pluto still exist. They come from an area called Kuiper's Belt.

a comet consists of three parts:

- A nucleus (mostly rock, ice and carbon)
- A head, or coma
- A tail (caused by heating and the reflection of the rays of the sun)

Tail of gas Tail of dust Nucleus Coma



Astronomers are honoured by having a comet named after them if they record such a space traveller for the first time. The comet Hale-Bopp, for example, got its name when Alan Hale and Thomas Bopp spotted and registered it at the same time. In 1997 this comet passed close to the earth and caused great excitement, especially since its nucleus was 40 km in diameter.

Comets travel in oval orbits around the sun. Sometimes when orbits cross, major collisions can occur.

Thus the planet Jupiter was bombarded by fragments from the comet Shoemaker-Levy 9 in July 1994. This was televised and millions of people watched the phenomenon.

(www.dustbunny.com/afk/planets/jupiter/jupimpact.htm)

Halley's comet was observed in South Africa in 1986. It takes 76 years to orbit the sun.

Comets probably contributed to the formation of our planet and probably also brought carbon and water to our planet.

Meteorites

- Particles that travel through space are called meteors. When large meteors hit planets, huge craters are caused. The craters on the moon were formed in this way. Some of the largest craters in South Africa are the one at Pretoria and the Tswaing crater, as well as the Vredefort crater in Northwest.
- Thousands of tons of small meteors usually hit the earth every day, but most are as small as a grain of sand. When they enter the earth's atmosphere, we call them METEORS, once they have hit the earth's surface they are called meteorites. Some are large enough to be seen with the naked eye and people call them shooting stars.
- Meteors as bright as Venus are called fireballs.
- The largest meteor that has ever hit the earth is known as Hoba. It hit Namibia and weighed about 60 tons. You can still see it near Grootfontein.

Asteroids

- Asteroids are giant meteors with a diameter of normally up to one kilometre. Approximately 40 000 asteroids revolve around the sun in orbits between the planets Mars and Jupiter, and for this reason this area is called the asteroid belt.
- Ceres is the largest asteroid and has a diameter of about 1 000 km.
- Many people believe that an enormous asteroid such as this hit the earth, creating dust clouds that blocked out all sunlight and causing dinosaurs to become extinct 65 million years ago.
- The last large asteroid hit the earth in Siberia in 1908.
- Scientists believe that the earth could be hit by an asteroid every few hundred years and so they keep a careful watch on the skies above us.
- Thousands of amateur astronomers around the world watch faithfully every night and report all "new" objects. These objects are referred to as "near earth objects" or NEOs.
- It is estimated that there are about a thousand NEO's in space relatively close to us and that only half of them have been discovered to date.

- Comets become visible only a few months before they present a danger to us.
- Marvin Kilgore, an American, currently has the largest collection of asteroid fragments and explains that asteroids have incredible mineral compositions that, if man had the technology, could be mined!

On the web:

Do visit the following websites:

neo.jpl.nasa.gov/neo.html

planet.dm.unipi.it/neodys/

impact.arc.nasa.gov

See how the skies above us are monitored by scientists!!

Activity: Questions on Comets and other Travellers in Space

1. What is:

- a) a comet?
- b) an asteroid?
- c) a shooting star?
- d) a crater?

2. Which comet:

- a) collided with Jupiter?
- b) was in our skies in 1986?

3. Where are examples of craters found in Southern Africa?

4. Where will you find:

a) Kuiper's Belt?

b) the Asteroid belt?

5. What is the name of the largest asteroid in the asteroid belt?

Assessment of QUESTIONS ON COMETS:

Did you answer the questions correctly?

[LO 2.4]

Assessment

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

2.4 applies knowledge.

Memorandum

COMETS AND OTHER TRAVELLERS THROUGH SPACE

ACTIVITY: QUESTIONS

1. a) Blocks of ice and rocks that are in orbit around the sun

b) giant meteorites

c) meteors

d) cavities formed by impact

2. a) Shoemaker- Levy 9

b) Halley

3. Tswaing in the vicinity of Pretoria, Vredefort Dome, Hoba in Namibia

4. a) Kuiper 's girdle – cold regions beyond Neptune and Pluto

b) between Mars and Jupiter

5. Ceres

Space Programme

NATURAL SCIENCES

Grade 9

THE EARTH ,GALAXIES AND SPACE PROGRAMMES

Module 17

SPACE PROGRAMME

- The space era for man began on 4 October 1957 with the launch of **Sputnik I** by the former Soviet Union.
- It consisted of a hollow metal ball that weighed only 84 kg and contained a radio transmitter and a thermometer.
- Today there is a launch of some kind of space craft almost every week.
- Konstantin Tsoilkovsky (1857-1935) was a teacher of mathematics who realised the possibilities of rockets, space stations and satellites a long time ago.
- With their **Apollo missions** the Americans made it possible for humans to walk on the moon. This happened on 20 July 1969 with the Apollo 11 mission. Neil Armstrong's words "One small step for man, one giant leap for mankind" are well known.
- Analyses and measurements of rock fragments have shed more light on our knowledge of our own moon.
- The "Near Shoemaker" mission, made it possible for the Americans to land a spacecraft successfully on Eros a large comet fragment. This resembles the dramatic representation in the film "Deep Impact").
- The US space shuttles have heralded a new era of technology with regard to computer programs, special metals and materials, satellite launches and much more.
- There have also been major disasters in the endeavours to master space. To date, two American space shuttles have exploded - first Challenger in January 1986 (during launching) and recently Columbia

on 1 February 2003. Columbia was the oldest shuttle and was on its 28th flight. It was the 113th shuttle flight and the first tragedy during a landing procedure.

- The South African Internet tycoon, **Mark Shuttleworth**, made headlines on 25 April 2002 by being the first person from Africa to travel in space.
- Thousands of South Africans and others visited his website daily and devoured the articles and news from space.
- His mission was to perform experiments on the International Space Station (ISS) and to gain valuable experience at the same time.
- Here is a brief summary of the mission:

African in Space

- On 25 April the Sojoes TM34 rocket was launched from the Russian space centre at Baikonor in Kazakhstan. On board were Mark, Joeri Gidzenko and the Italian Roberto Vittori.
- After a successful journey the shuttle reached the International Space Station (ISS) 2 days later.
- Mark and his companions spent 8 days on board the ISS and were watched by thousands of South Africans as the ISS crossed the country.
- Mark performed a number of important experiments on behalf of leading scientists and in this way obtained important research results and information.
- The team left the ISS on 5 May 2002 and returned to earth in a TM 33 Sojoes space craft.
- Mark and his companions landed the shuttle safely, its speed was slowed down by means of a parachute once it had entered the atmosphere and landed the space craft safely.

Activity: Mini-essay

Visit www.africaninspace.com and write a mini-essay on the value of the experiments in which Mark was involved in space.

Assessment of MINI-ESSAY:

Do you realise the value of man's discoveries and the value of the earth's sustainable resources? Do you appreciate all of these endeavours?

[LO 3.1; 3.2]

Satellites

- A satellite is an object that revolves around a planet. The moon is the earth's natural satellite.
- Man-made satellites are structures built by man and are usually launched by means of a rocket. This technology makes a wide variety of information possible - something that would not have been possible previously. Telecommunication satellites in particular have had a huge impact on our lives. Just think of how Telkom and your DSTV decoder work.
- The Hubble Telescope has given us unprecedented images of the universe, e.g. the impact of the Shoemaker-Levy comet that collided with Jupiter.
- Here in South Africa our own satellite, SUNSAT, was built by engineering students and academics of the University of Stellenbosch. It was launched on 23 February 1999 with the Delta 2 rocket, which also carried an American Air Force satellite and a Danish satellite. SUNSAT carries distance sensors and sends very high-resolution images back to us. Thousands of learners have been inspired by this project. (www.sunstep.sun.ac.za)

space stations

MIR

- The Russians launched the space station Mir in 1986. (Mir means peace and world.) It consisted of a nucleus of about 17 m in length. Sections such as Kvant 1 were gradually added and many people worked and lived there over the years.
- Many problems, such as wear and tear damage and power failures, were experienced with Mir and on 23 March 2001 it fell into the atmosphere and broke up into several pieces. People between Chile

and New Zealand were warned about the danger and it made news headlines all over the world.

- Do visit www.cnn.com/SPECIALS/2001/mir/

The ISS

- The ISS or International Space Station is a giant earth satellite that orbits around the earth at a height of 355 km and at a speed of 27 000 km/h. At this stage many of the planned missions has been delayed due to unforeseen problems.
- A number of countries are assisting in extending the ISS bit by bit. It is a perfect example of international cooperation between different countries. The idea is to complete it by 2006 at a proposed cost of 370 billion dollars.
- Laboratories inside the ISS will enable scientists to use the environment in which there is no gravity, for research on:
 - chemical reactions;
 - agricultural products;
 - cultivating cells;
 - medicine against AIDS and other diseases;
 - new metal alloys;
 - warning systems against giant asteroids, and
 - a halfway-station between the earth and the rest of our solar system.

SPACE PROBES

- Space probes are launched to investigate other planets and areas of the solar system.
- The Viking series probed planets such as Mars and landed on Mars in 1976. A smaller spacecraft, the Pathfinder, landed on Mars in 1997 and carried out investigations with a robot vehicle, the Sojourner.
- Pioneer 10 and 11 were launched further into space and are already more than 11 billion kilometres from the earth.
- Voyager 2 has passed Saturn, Uranus and Neptune.
- The Cassini-Huygens research craft reached Saturn in July 2004. The Cassini orbiter is orbiting Saturn, while the Huygen craft landed on Saturn's moon, Titan.

- In April 2003 the British launched the Beagle II with a Mars Express unit that arrived on Mars by Christmas 2003.

So many nations - So many missions!

Why not coordinate all the efforts?

- The reason: for having so many space probes is that the more efforts there are, the greater the degree of success. Many space missions do not succeed owing to problems with launching, orbiting or landing. NASA will attempt 3 landings on Mars within a period of 15 months.
- After the explosion of the shuttle Columbia, questions were again asked about the value of space research. The conclusion, in the words of Bill Readdy, a NASA administrator at the time of the Columbia tragedy, was: “Pushing the frontiers of space is dangerous”, and in those of President George Bush: “Our journey into space will go on...”

DEEP SPACE

- This is where stars are born and where so-called black holes occur.

Life cycle of a STAR

- A star starts as a gas cloud.
- This gas cloud heats up as a result of the force of gravity.
- A nuclear reaction starts and a star is born.
- As nuclei are converted, the star becomes hotter.
- The nucleus becomes increasingly hotter and the surface cools down.
- The star expands and is now called a red giant.
- Layers are discarded, the inner core collapses and a white dwarf is formed.
- This will happen to our sun in about 5 billion years' time!
- When a star dies, it explodes in a supernova - this is as bright as a million suns!
- The section that remains can be either a neutron star or a pulsar or a black hole.
- A black hole is formed when matter is compressed until it is so dense that not even light can escape the pull of its gravity.

Colours of STARS

- The whiter the star, the hotter it is and the brighter it shines.
- The redder it is, the cooler and fainter it is.
- Red stars that are bright are therefore closer and white stars that are faint are very far away.

Bright STARS in SPACE

- If you are in the Southern Hemisphere and look south (i.e. the sun has set to your right), you will see two bright stars. They are α -Cen, also known as Alpha Centauri, and β -cen or Beta Centauri. Together they are called the Pointers, as they point to the Southern Cross.
- Excluding the sun, α -Cen is our nearest star.
- Let's see whether it's possible to send a shuttle to α -Cen.

Description	Calculation
<i>Earth to α -Cen</i>	4,26 light-yearstherefore: $4,26 \times 9,5 \times 10^{12}$ km= 10 rounded off
Shuttle speed	10 km.s^{-1}
<i>Time to α -Cen</i>	$4 \times 10^{13} / 10 = 4 \times 10^{12} \text{ sec}$ (Speed = distance / time)
How long is this in a human lifespan of 70 years?	$70 \times 365 \times 24 \times 3\,600 \text{ sec} = 2,2 \times 10^9$ sectherefore: $4 \times 10^{12} \text{ sec} / 2,2 \times 10^9$ sec1812 years
Therefore it is fairly impossible today!!!	

Activity: DEEP SPACE

1. Why should man have space programmes and launch space probes?
2. Compile a flow diagram of the phases of a star's life cycle.
3. State two ways in which satellites currently play a role in your life.
4. Explain the following astronomical terms:

Red giant	
Black hole	
Space probe	
White dwarf	
Supernova	
Pulsar	

5. State the benefit of the ISS for mankind.

Assessment of QUESTIONS ON DEEP SPACE:

- a) Do you understand the terms?

b) Could you do the flow diagram correctly?

c) Do you understand the reasons for space programmes?

[LO 2.1; LO 2.2; LO 3.1]

Assessment

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;
- categorises information.

LO 3: Science, Society and the Environment

The learner will be able to demonstrate an understanding of the interrelationships between science and technology, society and the environment.

This is evident when the learner:

3.1 understands science as a human endeavour;

3.2 understands sustainable use of the earth's resources.

Memorandum

SPACE PROGRAMMES

ACTIVITY: QUESTIONS

1. to gain more knowledge of possibilities beyond our own planet

2. gas cloud heating forces nuclear reaction expansion layers cast off
inner core collapses explosion/supernova neutron or “black hole” remains

3. telecommunication, television transmissions, weather forecasts

4.

Red giant	Stage of exansion in the life cycle of a star
Black hole	Stage of deterioration at the end of a star's life
Space probe	Investigations of space
White dwarf	Collapse of a star
Supernova	Final explosion results in clouds of gas
Pulsar	Neutron star or remains

5.	• Chemical reactions. • Agricultural products.
----	--

- Cultivation of cells.
- Medicine against AIDS and other illnesses.
- New alloys of metal.
- Warning systems to alert against giant asteroids.
- Possibly a future "half-way house" between the earth and the rest of the solar system.

Energy

NATURAL SCIENCES

Grade 9

ENERGY, FORCES AND MACHINES

Module 18

ENERGY

Activity:

Naming, discussing and identifying energy sources and different forms of energy

[LO 2.2; LO 2.3; LO 2.4]

What is energy?

- Energy is the ability to do work: to make things happen and to bring about change.

Forms of energy:

- Energy occurs in different forms, e.g. as sound, warmth, light, chemical compounds and electricity.
- Some objects have energy because they have motion, or simply because of their position. This has to do with *kinetic energy*(motion)

or *potential energy* (position).

- Energy cannot be seen or felt through touch, but is present everywhere: in the surrounding air, in each piece of matter, in each of the cells that make up our bodies. Energy is the only thing that existed before the big bang that marked the origin of the universe!

Sources of energy

- Energy in **ELECTROMAGNETIC** form plays an important role in technological innovations like radio transmission towers, microwave ovens and X-ray machines. Scanners are machines that reveal images of things that are normally invisible, e.g. the internal organs of the body. The scanner sends sound waves into the body. When the patterns formed by the waves change, it shows an image that is of value in helping medical practitioners to make a diagnosis. The image that is produced is known as an MRI or Magnetic Resonance Image. This process is invaluable in the field of medicine.
- We can distinguish three important sources of energy, namely:
 - the sun
 - atomic nuclei
 - the movement of the earth and its gravity, and the effect that other planets have on the earth.
- **Solar energy** originates from the collision and fusion of atoms (nuclear fusion). This energy reaches us in the form of light and warmth. Heat energy causes atmospheric currents that result in wind, another important potential source of energy. Heat energy also results in evaporation of water, which leads to the formation of clouds which bring rain. When rain falls in high-lying areas, rivers are formed, yet another important source of energy. Solar energy is also captured in foodstuffs when photosynthesis occurs in plants. The food we eat and the fuel we use (e.g. petroleum) are derived from this process.
- **Nuclear energy** is obtained when people use advanced technology to split atoms. Nuclear power stations convert nuclear energy into electrical energy.
- The **movement of the earth and its gravity, together with the influence that other planets exert on the earth**, work together to

create ocean currents and to cause the downward flow of rivers. Such energy can be utilised in different ways, e.g. for transportation or for driving machines, like turbines.

Try to identify the form of energy illustrated by each of the following examples and explain the sources of such energy.

Example of energy
• a book lying partially overhanging the edge of a table
• a cellular network sending sound waves to your cell phone
• a ski-jumper jumping off a ski slope
• an express train racing across the plains
• heating your lunch in a microwave oven
• a doctor taking x-ray photographs of your broken hand

• a robot in a car manufacturing plant
• a lightning bolt and a thunderclap
• the energy in your lunchtime meal
• sound that causes the eardrum to vibrate
• electrical energy at Koeberg
• the light of a torch

[LO 2.2; LO 2.3; LO 2.4]

Activity

Illustrating the law of energy conservation by means of different examples of energy conversion

[LO 2.4]

Conservation of energy

- Herman Helmholtz (1821 – 1894) developed the law of ENERGY CONSERVATION.
- This law states that *ENERGY CANNOT BE CREATED OR DESTROYED*. It can only be converted from one form to another.

Indicate the form of conversion that is relevant in each of the following instances:

- A ball rolling from the top of a hill
- A piece of wood burning
- The ESKOM power stations
- Pumping up your bicycle wheel
- Sun panels attached to a satellite
- A boy eating an ice-cream
- A torch shining
- A modern submarine travelling along the coastline
- A person skating on ice
- A water wheel
- A pistol shot
- A glowing electric light bulb

Write critical comments on the following statement: *A steam train runs on solar energy.*

Assessment of insight into the conservation of energy

Could you provide examples of the law of energy conservation?

[LO 2.4]

Assessment

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- categorises information;
- interprets information;

2.4 applies knowledge.

Memorandum

ENERGY

Activity

Forms of energy illustrated:

- Mill: Kinetic
- Atomic bomb: Nuclear/atomic
- Radio tower: Radiation
- Heater: Radiation
- Battery: Chemical
- Light bulb: Radiation
- Food: Chemical

Table:

Example	Form	Source
Book on table	• Potential	Food supplied chemical energy by which the book was lifted onto the table through kinetic energy (the action of the muscles)
Ring of cell phone	• Radiation	Chemical or nuclear: electricity
Ski-jumper	• Potential and kinetic	Chemical (food or electricity) transported the person to a higher point
Express train	• Kinetic / mechanical	Chemical or nuclear: electricity
Microwave oven	• Radiation	Chemical or nuclear: electricity
X-ray plates	• Radiation	Chemical or nuclear: electricity
Factory robot	• Kinetic / mechanical	Chemical or nuclear: electricity
Lightning bolt	• Radiation (warmth,	Chemical: electricity

	light, sound)	
Lunch	• Potential, chemical	Chemical: sunlight that is held (in the food)
Sound	• Radiation	Chemical: electricity
Electrical energy at Koeberg	• Chemical / electrical	Atoms
Light from a torch	• Radiation (warmth, light)	Chemical

Activity

- Ball: potential to kinetic and sound
- Wood: chemical / potential to sound, light, heat
- Eskom: chemical / potential to electrical
- Bicycle wheel: chemical / potential to kinetic/mechanical
- Satellite: radiation to electrical
- Ice-cream: chemical / potential to chemical / potential (stored in the body) to kinetic
- Torch: chemical / potential to light and heat
- Submarine: atomic / nuclear / chemical to electrical to kinetic
- Skating: chemical / potential to kinetic
- Water wheel: kinetic to electrical
- Pistol: chemical / potential to kinetic, heat and sound

- Light bulb: electrical to light and heat
- Yes: Steam is derived from coal, which is dead plant material formed means of photosynthesis, with the help of sunlight (radiation by energy).

Forces and friction

NATURAL SCIENCES

Grade 9

ENERGY, FORCES AND MACHINES

Module 19

FORCES AND FRICTION

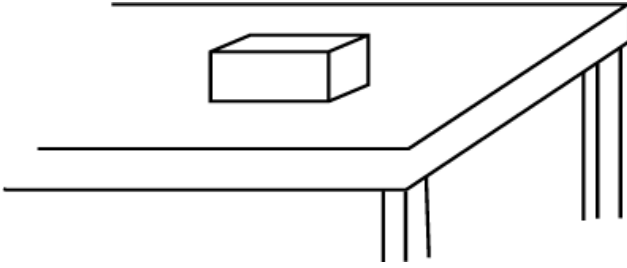
Activity:

Describing and applying the terms FORCE and FRICTION

[LO 1.2; LO 1.3; LO 2.3; LO 2.4]

The sketch that follows represents a brick lying on a table in front of you. Discuss ways in which the brick could be made to move and decide on one method. In science we indicate forces and the direction in which they operate by means of an arrow.

Indicate the possible directions of motion by means of arrows.



- How would you define a force?
- You could have applied tractive power (pull) or thrust power. When you stop pulling or pushing the brick, it stops moving or comes to rest. Why? Another simultaneous force that opposed the force that you were applying must have been in operation. This force is known as **FRICTION**.

List examples of friction from common daily experiences.

Assessment

Were you able to describe the term “force”, indicate the lines of action on the sketch and provide examples of friction?

[LO 2.4]

Investigation: FORCES AND FRICTION

Use a stopwatch, a spring balance, a brick, a smooth piece of cardboard or perspex as big as the base of the brick, a length of string, some baby-powder and a skateboard.

- Tie the brick to the spring balance, using the length of string.
- Move the spring balance along slowly to bring the brick into motion, checking the balance to note the position of the needle when the brick begins to move.
- Pull the brick over a distance of 5 metres – get someone to time this action and enter the information on a table.
- Repeat this procedure, but secure the brick to the cardboard or perspex. Time the action and tabulate the details. Try to apply sufficient force

for the needle to give the same reading as previously. Use same distance stretch for every procedure.

- Repeat the procedure, but secure the brick to the skateboard.
- Do three repeats of all of the above. Use baby powder as an alternative in all three and sprinkle on wheels of skateboard.

1. Why was it necessary to repeat the experiment three times?
2. Why did you get different time readings for the action when you pulled the brick only, the brick on cardboard or perspex and the brick on the skateboard?
3. Mention other methods by which friction can be reduced.
4. Think about a motor vehicle and list five examples to show that it is necessary to use a lubricant or some other method designed to reduce friction. Explain each example.

[LO 1.2; LO 1.3; LO 2.3]

Remember the following:

- Forces can:
 - change the shape of an object;
 - increase or slow down the speed of an object;
 - redirect the motion of an object and
 - rotate an object around a fixed point.
- There are *forces of attraction or pull (tractive forces) and of thrust (push)* and can both operate directly or over a distance.
- Most forces, however, only come into operation when objects are in contact with one another.

Questions:

1. Provide an example of each of the following effects:
 - change of form

- change of speed
- forces that operate when there is no contact
- forces of thrust

Activity:



Researching a scientist who made an important contribution to science:
Isaac Newton

[LO 1.1; LO 1.3; LO 3.1]

Write a report (three A4 sheets) on Isaac Newton and on his contribution to science. Also refer to the saying 'The apple never falls far from the tree'. Include your report in the module as extra pages following this one.

Assessment of RESEARCH

Were you able to find out who Isaac Newton was and do you understand his contribution to science?

[LO 1.1; LO 1.3; LO 3.1]

Activity:

Describing the relationship between mass, weight and force

[LO 2.3; LO 2.4]

The earth attracts all objects that are in its proximity.

- We know the following:
 - all objects have mass and all masses exert tractive force (pull); this mostly is not observable;
 - the greater the masses, the greater the tractive force; as the earth is large, we only observe tractive force in one direction (i.e. as the earth attracting our bodies);
 - the closer the proximity between masses, the greater the tractive force between them.
- The *mass* of an object serves as an indication of the *amount* of matter and this is measured in
- The *weight* of an object signifies the amount of tractive force (pull) that the earth exerts on the particular object.
- The tractive force which attracts objects to the earth is known as *gravitation*.
- The unit for measuring force is **NEWTON (N)**.

The weight of an object that has a mass of 100 g is 1 N

- This is roughly equal to the weight of a small banana or a small apple.

Questions:

1. ONE kilogram has a weight ofN.
2. Your mass will remain the same, no matter where you may be in the universe, but your weight will vary. Explain this. (*Refer to the module dealing with space programmes.*)
3. What is your mass?.....
4. What is your weight?.....
5. What would your weight be on the moon where the tractive force is 0,6 N?
6. FORCE and WEIGHT are both measured in NEWTON. Explain this.
7. You are able to observe the effect exerted on an object by the force of attraction if you lift it up by means of an elastic band. This demonstrates the principle behind the **SPRING BALANCE**.
8. Describe the difference between a spring balance and a bathroom scale.

[LO 2.1; LO 2.3; LO 2.4]

Activity:

Measuring force (weight) with a spring balance

[LO 1.2; LO 2.4]

Measure the weight of a number of objects with the help of a calibrated spring balance.

- First measure the weight after suspending the object from the spring balance by means of a length of string.

- Then place the object on the floor and pull it along to get a reading. Note down both readings.
- Repeat each operation three times for each object.

Assessment of the ability to measure weight/force by means of a spring balance

[LO 1.2]

QUESTIONS

1. What is inside the spring balance to make it possible to take a measurement?
2. Why do the two readings differ?
3. What is the meaning of *calibrated*?
4. What would the weight of a 2,5-kg bag of sugar be?
5. What is the purpose of the adjustment for zero on the spring balance?
6. Identify the forces that are present.
7. Distinguish between weight and mass.

WEIGHT

MASS

Assessment of QUESTIONS on measuring force/weight

Were you able to apply your KNOWLEDGE?

[LO 2.4]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- plans investigations;
- conducts investigations and collects data;
- evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;
- interprets information;

2.4 applies knowledge.

LO 3: Science, Society and the Environment

The learner will be able to demonstrate an understanding of the interrelationships between science and technology, society and the environment.

This is evident when the learner:

3.1 is able to show an appreciation of science as a human endeavour.

Memorandum

FORCES AND FRICTION

Activity

Directions of motion: to the left, right or upwards.

A force: the ability to do work; by pushing (thrust) or by pulling; friction is another example of a force.

Examples of friction: learners could name a wide range of examples, e.g. the rubber from car tyres on tar, engine parts that are not lubricated, a chair on the floor, soles of shoes against the floor, etc.

Table: accept the times provided by learners.

Questions (first set):

1. To obtain an average that would supply a more accurate indication, as conditions may not be always be exactly the same.
2. The friction is reduced: bricks alone provide the most, cardboard / perspex causes less (because of the smoother surface), a skateboard involves the least friction.
3. Lubrication, objects like small pellets that fill the space between the two surfaces reduce the surface of friction, etc.
4. Ball bearings, pistons in cylinders, toothed wheels, chains, door hinges, steering mechanism, springs, speedometer cable, a jack, etc.

Activity

Assist the learners by means of a framework according to which the report is to be assessed:

- Planning (stating the problem, identifying sources)
- Execution (collecting data)
- Evaluating and communicating data
- Appreciation of science as a human endeavour

Activity

Mass: measured in gram or kilogram

Questions:

1. 10N
2. Weight is determined by means of the attractive force of the earth, which may vary from one place to another. Mass remains unchanged.
3. Learner's answer
4. Learner's mass in kg $\times$ 10N
5. Mass $\times$ 0,6N
6. Weight signifies the force with which the earth attracts the body. It therefore has to be measured in the same unit.
7. –
8. The spring balance directly measures the earth's attractive force. A balance (pair of scales) measures by comparing an object with a known mass ('weights') or by using a calibrated resistance (e.g. a spring).

Activity

Questions:

1. A spring
2. Friction is added to the pull
3. Marked according to acknowledged units of measure, e.g. metric units
4. 2,5 Kg $\times$ 10N = 25 Kg
5. It makes it possible to readjust the spring balance.
6. Gravitation and friction

1. Weight: attractive force of the earth on mass

Mass: the amount of matter

Work

NATURAL SCIENCES

Grade 9

ENERGY, FORCES AND MACHINES

Module 20

WORK

Activity:

Calculating work/effort (performance)

[LO 2.3; LO 2.4]

We have established that objects can be moved around by applying FORCES. The application of force means that WORK is performed. **WORK** is the scientific term used to refer to this application of force.

We say that 1 Joule of work is required to move the force of 1 Newton over a distance of 1 metre!

- Remember that effort actually indicates the use of energy. A Joule therefore indicates the amount of energy that is required to move 1N over a distance of 1 metre.

Questions:

1. How much effort do you apply when you:

a) drag a 10-N brick over a distance of 3 metres?

b) drag a brick with a mass of 2,5 kg over a distance of 5 m?

2. The energy value of a bar of chocolate is 300 **kJ**. How many 2,5-kg bricks will you be able to drag for a distance of 5 metres with this amount of energy?

3. How far will the energy provided by another bar of chocolate enable you to drag one such brick?

You have made use of the formula :

$$W = F \times s$$

W : Work **F** : Force **s** : Distance

- **WFs**The placement of the letters in the triangle will help you to remember how to use the formula. Your educator will explain this to you.
- Now calculate how much energy will be required to move yourself over a distance of 10 metres.

Your educator will provide you with further exercises dealing with movement/acceleration.

Assessment of **CALCULATION OF ENERGY USAGE (EFFORT)**

[LO 2.3; LO 2.4]

Assessment

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- interprets information;

2.4 applies knowledge.

Memorandum

WORK/EFFORT

Activity

Questions:

1. a) $F \times s = 10\text{N} \times 3\text{m} = 30 \text{ Joule}$

b) $F \times s = (2,5 \times 10) \times 5 = 125\text{J}$

2. $300 \text{ kJ} = 300\,000\text{J}$ divided by $125\text{J} = 2\,400$ bricks

3. $300\,000\text{J}$ divided by $25\text{N} = 12\,000 \text{ m} = 12 \text{ km}$

- Energy required to move a learner over a distance of 10 m:

Learner's mass = A

$$W \text{ (energy)} = F \text{ (force)} \times s \text{ (distance)} = (A \times 10) \times 10\text{m}$$

Work and conversion of energy

NATURAL SCIENCES

Grade 9

ENERGY: FORCES AND MACHINES

Module 21

WORK AND CONVERSION OF ENERGY

Activity 1:

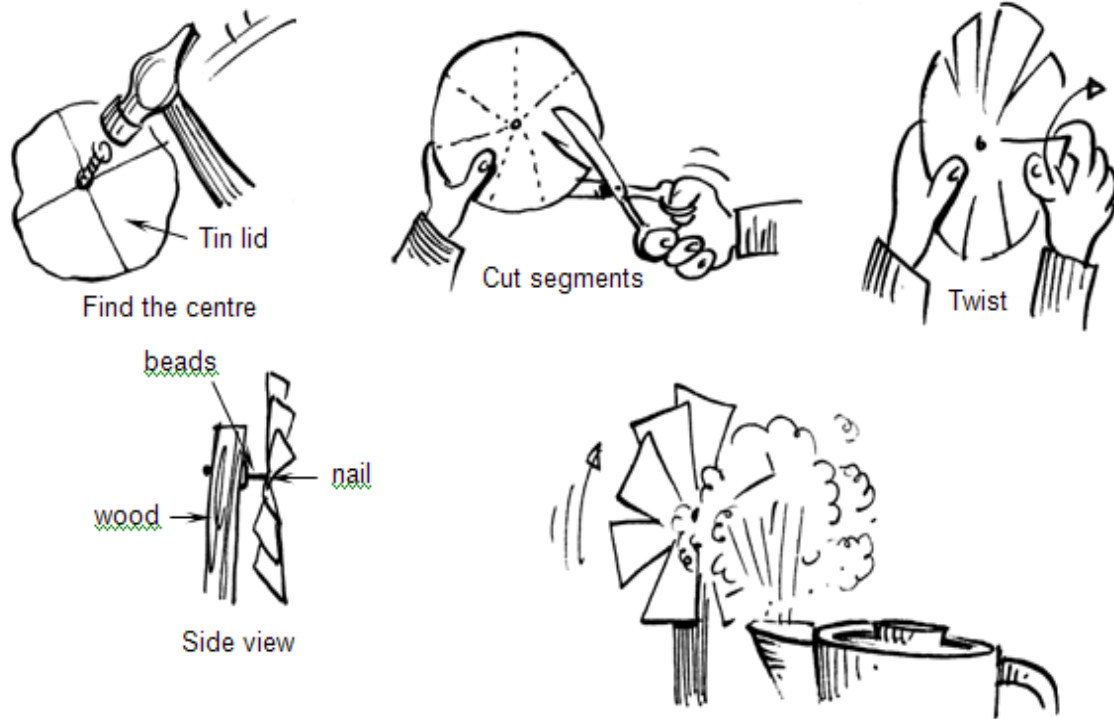
Describing the relationship between effort and energy with reference to a turbine

[LO 1.2, 2.1, 3.2]

Energy is necessary for any performance of work. This energy must be obtained from FUELS or from CONVERTED forms.

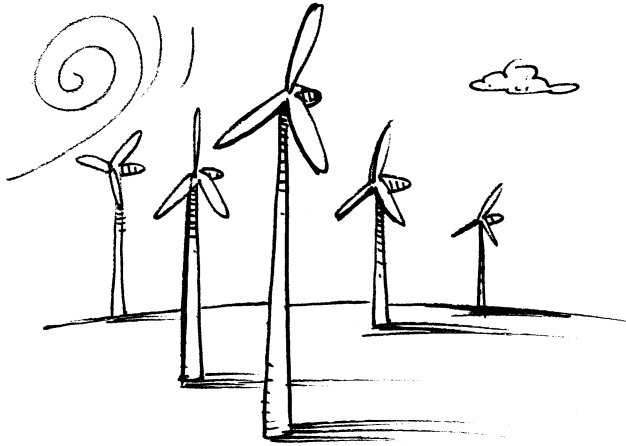
CONSTRUCTING A SIMPLE TURBINE

- Use a tin lid, cut it into segments to form it into a fan and make a hole in the centre.
- Nail the fan to a piece of wood.
- Boil water in a kettle, until there is a strong draught of steam.
- Hold the fan in position so that the jet of steam will be directed towards its centre.



- QUESTIONS

1. What happened?
2. What kind of energy conversion occurred?
3. Describe the form of energy conversion that occurs with wind towers such as those that have recently been erected along the West Coast in the vicinity of Malmesbury.



4. What advantages do wind towers like these have for the environment?

- Write a paragraph (60 words) about the value that the principle of the turbine has for people.
- **WORK** is also measured in (give the name of the unit)
.....

Activity 2

Discussing other machines that operate by means of conversion of energy

THE MICROSCOPE

- As long ago as the early 17th century, Robert Hook assumed that light rays have energy and that a wound-up spring has potential energy. He also constructed the first microscope and was the first person to use the universal lever, which is used in a wide range of instruments and cars.

SATELLITES

- Satellites depend on solar panels to absorb light energy from the sun and supply electricity for all other electronic functions and for cameras and radio equipment. Energy is also stored in chemical form in rechargeable batteries.

WIND PUMPS

- The blades of the rotary disk are turned by the wind to drive a pump that pumps underground water to a storage dam.

Assignment 3: GROUP DISCUSSION

- Name at least three other machines that operate by means of energy conversion and explain the workings of each one.

WHEN IS A MACHINE MECHANICALLY INEFFECTIVE?

- When a machine or appliance fails to provide an adequate quantity of the energy that it has been designed to provide we say that it is ineffective.
- The light bulb, for instance, releases (unwanted) heat, but was actually designed to provide light!
- In this instance, fluorescent lighting is more effective and economical.
- Are you able to list further examples?

Assessment

LO 2
Scientific KnowledgeThe learner will know and be able to interpret and apply scientific, technological and environmental knowledge.
This is displayed when the learner:
2.1 is able to recall meaningful information;
2.2 is able to arrange information according to categories;

2.3 is able to interpret information;
2.4 is able to apply knowledge.
LO 3
Science, Environment and the Community The learner will be able to demonstrate an understanding of the interrelationships between science and technology, society and the environment.
This is displayed when the learner:
3.1 is able to show appreciation of science as a human endeavour;
3.2 is able to recognise the value of sustainable utilisation of resources.

Memorandum

Activity 1

Questions:

1. The fan ('turbine') rotates
2. Chemical to steam to kinetic / mechanical
3. Kinetic to electrical to chemical
4. Natural resources (e.g. coal) are not used and pollution is avoided

Paragraph:

- E.g. natural energy is converted to usable energy
- Without consumption of resources
- Without pollution
- There are no waste products to be got rid of

- Examples of utilising turbines: generation of electricity, previously also for driving mills

Effort is measured in Joule

Activity 2

- Any power generator
 - Compressor
 - Vehicle engine
 - Sewing machine
 - Kitchen appliances
 - Etc.
-
- Ineffective machines: any machine or system that releases energy in a non-usable form, e.g. a car that is driven with the hand break on (much energy is wasted in the attempt to overcome the force of friction – energy in the form of heat goes to waste).

Power

NATURAL SCIENCES

Grade 9

ENERGY, FORCES AND MACHINES

Module 22

POWER

Performance (effort/work) is important, but the SPEED at which it is performed is also important.

Activity:

Explaining the concept POWER and calculating it

[LO 2.3; LO 2.4]

POWER = WORK / TIME

We measure this in J/s, or Watt (W)

1J per second provides 1 Watt of power

- The term **Watt** is derived from the name of James Watt, who was responsible for improvements to the steam engine.
- The electricity in your home is measured in kilowatt-hour (**kWh**).
- One kilowatt-hour is the energy output of a 1 kW appliance in one hour.

- The longer such an appliance is operated, the more energy is used. A 2 kW heater that is operated for three hours will therefore use 6 kW energy.
- Complete: kW per hour x hour = 6 kW.
- The cost of a kWh varies but comes to approximately R0,35 in the Western Cape.

Assignment:

Measure your own power capacity

- Identify a stairway in the school and measure the height of a step.
- Multiply this by the number of steps.
- Determine your weight.
- Run up the stairs and take your time in seconds.

Complete the following table:

Aspect	Measurement
Step height	
Mass	
Weight	
Work performed: $F \times s$	
Time	

QUESTIONS

1. What was your power?
2. Explain the following:
 - a) Kilowatt
 - b) Kilowatt-hour
3. Find out what the power rating of the average car (your educator's car, for instance) is.
4. Would you be prepared to push your educator's car all the way home? Why not?
5. Why would it be easier for the car engine to do this, even though you might have enough time and help from your friends?
6. Calculate the cost of operating a 3 000 W heater for a month (July) for 4 hours per day at a tariff of R0,35 per kWh.

Assessment of the CALCULATION of POWER CAPACITY

Were you able to apply knowledge and do the calculations?

[LO 2.3; LO 2.4]

Activity:

Doing RESEARCH on power and performing relevant calculations

[LO 1.1; LO 1.2; LO 2.2; LO 2.4]

Find out what 1 kWh of electricity costs in your municipal area.

Suppose you use the following appliances during the day:

- Computer for 2 hours
 - Television for 3 hours
 - Lamp for 4 hours
 - Kettle for 15 minutes
 - Stove for 45 minutes
 - Hairdryer for 15 minutes
- Determine the average power of each of these items by reading the specifications given on the appliances. Complete the table.

Electrical appliance	Power (kW)
Computer	
Television	
Lamp	
Kettle	
Stove	
Hairdryer	

Now complete the following cost assessment:

APPLIANCE	POWER (kW) as above	TIME Usage(hours)	Energy Usage (kW/h)	Cost @ R1,50	
-----------	---------------------------	----------------------	---------------------------	--------------------	--

				per hour	
Computer					
Television					
Lamp					
Kettle					
Stove					
Hairdryer					

QUESTIONS

1. What is your deduction?
2. Are you aware of the need to use energy more economically? Give your own opinion of this.
3. Calculate the following:
 - a) A mountaineer whose mass is 75 kg climbs a 320 m rock face in 20 minutes. Calculate his power.
 - b) A large cupboard weighing 150 kg has to be carried up a stairway approximately 5 m high during a move.
 - How much work is performed?
 - What is the performance rating of the workers if two men are able to do this in 10 minutes?

Assessment of RESEARCH on appliances

Were you able to obtain all the required information and do the calculations?

Could you categorise information?

[LO 1.1; LO 1.2; LO 2.2; LO 2.4]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- plans investigations;
- conducts investigations and collects data.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- categorises information;
- interprets information;

2.4 applies knowledge.

Memorandum

POWER

Activity

Questions:

Power = Labour (work) / Time

Labour = Power (F) × s

Weight = Mass × 10 = Kg

$$= \underline{F \times s} \text{ Time} =$$

2. a) 1000 Watt, in other words, $1\,000 \times \text{Joule/second}$

b) The amount of kW that is delivered or consumed in one hour

3. Larger cars approximately 100 kW; smaller cars approximately 70 kW

4. –

5. Engine has greater power / capacity for doing work

6. First calculate total consumption of energy (kWh):

$$= \text{kW} \times \text{h (hour)} = 3 \text{ kW} \times 124 \text{ h} = 372 \text{ Kw}$$

Then calculate the costs:

$$= 372 \times \text{R}0,35 = \text{R}130,20$$

Activity

The power per hour of appliances (more or less):

- Pocket calculator 0,0006W
- Microwave oven 750W
- Drill 500W
- Radio 28W
- Toaster 800W
- Fan 100W
- Iron 1 200W
- Stove 1000W
- Computer 250W

Questions:

1. The greater the power, the higher the cost of using it

2. Debate

$$3. \text{ a) } W = F \times s = (75 \times 10) \times 320 = 750 \times 320 = 240\,000 \text{ J}$$

$$\begin{aligned}
 \text{Power} &= \frac{240000}{(20 \times 60)} \\
 &= \frac{240000}{1200} \\
 &= 200 \quad (\text{or } 0,2 \text{ kW})
 \end{aligned}$$

$$b) W = F \times s = (150 \times 10) \times 5 = 7500 \text{ J}$$

Equation:

$$\begin{aligned}
 \text{Drywing} &= \frac{\text{Arbeid}}{\text{Tyd}} \\
 &= \frac{7500}{600} \\
 &= 12,5
 \end{aligned}$$

$$\text{One worker's performance} = 12,5 \text{ J/s} = 12,5 \text{ W}$$

Machines make work easier

NATURAL SCIENCES

Grade 9

ENERGY, FORCES AND MACHINES

Module 23

MACHINES MAKE WORK EASIER

Activity:

Comparing a variety of APPLIANCES

[LO 1.3; LO 2.2; LO 2.3]

A machine is a technologically designed apparatus that can perform work through the conversion of energy from one form to another.

- Energy makes work possible.
- Work involves transfer of energy.
- The amount of energy that is applied is equal to the performance that is provided.

Small machines have the capacity to effectively provide the power that human beings are simply unable to provide.

When you buy small electrical appliances that have to perform work, you ought to check the kilowatt power. By doing this you will be able to determine whether you will obtain the required performance.

Case Study

Comparing APPLIANCES

- You want to buy a hairdryer as a birthday present for your sister and a kettle to replace your mother's kettle, which has stopped working.
- Your sister gets up late, but washes her hair every day. She spends 20 minutes blow-drying it every morning.
- Your mother regularly makes tea for the family and for friends who come to visit. This comes to at least four cups of tea a time, at least five times per day.

Compare appliance models of your choice and prices at a wholesaler. Also investigate performance and make a suggestion concerning what you are able to buy.

- Write a brief report on your findings.

Assessment of CASE STUDY regarding appliances

Were you able to interpret the information and apply knowledge?

[LO 1.3; LO 2.2; LO 2.3]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

1.3evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;
- categorises information;
- interprets information.

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Levers

NATURAL SCIENCES

Grade 9

ENERGY, FORCES AND MACHINES

Module 24

LEVERS

Activity:

Identifying levers as mechanisms to identify energy transfer

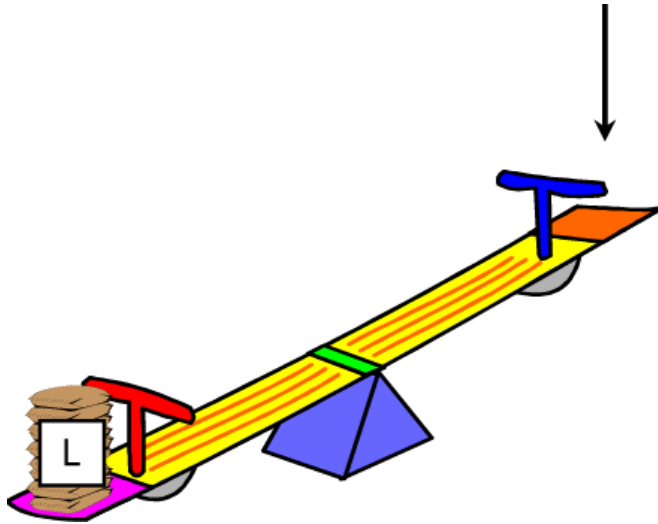
[LO 2.2; LO 2.3; LO 2.4]

The operation of the simple appliances given above depends on the principle of the lever.

- All levers have a **PIVOTAL POINT** or **FULCRUM**, a point where effort or power is applied (or the **POWER POINT**), and an **AREA OF LOAD** called **LOAD (L)**.
- The relevant symbols are:
 - P = point/fulcrum
 - E = effort/power
 - L = load
- Levers are classified according to the position of the fulcrum.
- There are three classes of levers.

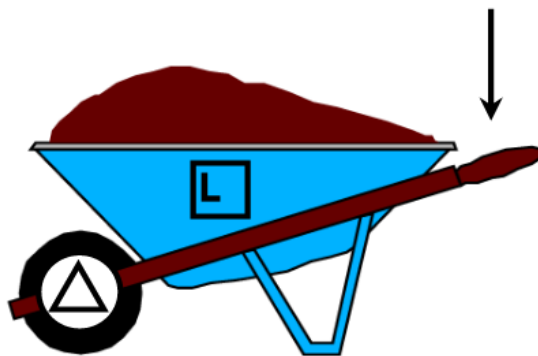
- Class 1 lever
- Class 2 lever
- Class 3 lever

CLASS 1: Fulcrum in the middle



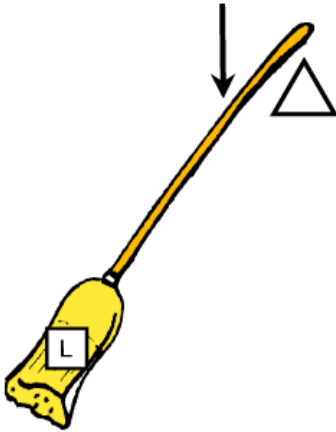
e.g. a see-saw

CLASS 2: Load in the middle



e.g. a wheelbarrow being pushed

CLASS 3: Effort/Power in the middle



e.g. a golf club or broom held for use

Identify the class of lever in each of the following examples:

- a) You pick up a can of cold drink and raise it to your mouth.
- b) You stand on the tips of your toes.
- c) You hold a fishing rod, having hooked a large fish.
- d) You row a boat using an oar.

Assessment of classification of LEVERS

Were you able to place the examples in categories?

[LO 2.2; LO 2.3]

Activity:

Calculating mechanical advantage

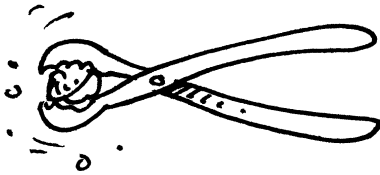
[LO 2.4]

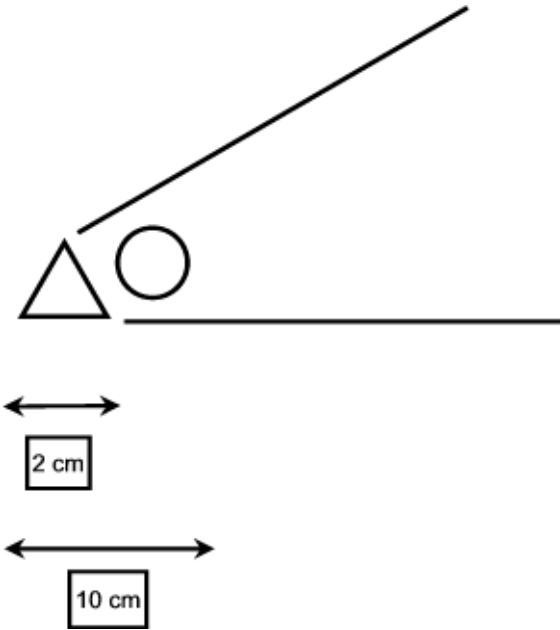
Force $\times$ distance = load $\times$ distance

- It is possible to calculate the mechanical advantage of a lever: the ratio of **load** to **force** provides this index.
- We refer to this as *mechanical advantage*.

1. How would one be able to enhance the mechanical advantage of a class 1 lever?

1. You have a nutcracker with the following dimensions:





- You require 150 N to crack the nut.

- a) Which FORCE is needed?
- b) What is the mechanical advantage of the nutcracker?

[LO 2.4]

The Wheel

Activity:

RESEARCHING the invention of the WHEEL and evaluating as an example of human endeavour

[LO 1.1; LO 1.2; LO 2.1]

The invention of the wheel more than five thousand years ago resulted in a remarkable change in human performance of work.

The wheel is like a lever, except that the movement comprises rotation around an axle and not an up and down movement around a fulcrum.

Project dealing with the WHEEL

Work as a group and create a poster to represent the development of the wheel and to illustrate different ways of using wheels.

- Did the Egyptians have wheels to assist them when they were building the pyramids?
- How did they manage to perform all these tasks?

Assessment of RESEARCH on the WHEEL

Were you able to plan and execute the investigation and to evaluate it? Were you able to gain INSIGHT into the VALUE of human scientific endeavour?

[LO 1.1; LO 1.3; LO 3.1]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- plans investigations;
- conducts investigations and collects data;
- evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;
- categorises information;
- interprets information;

2.4 applies knowledge.

LO 3: Science, Society and the Environment

The learner will be able to demonstrate an understanding of the interrelationships between science and technology, society and the environment.

This is evident when the learner:

3.1 is able to show an appreciation of science as a human endeavour.

Memorandum

Activity

- a) Class 3
- b) Class 2
- c) Class 3
- d) Class 1

Activity

L.W. A study of levers is no longer required as part of the syllabus, but levers are referred to in this module *to illustrate the transfer of energy from one body to another.*

1. Reduce the force distance

2. a)

b)

Activity

Assess according to the learning outcomes and assessment standards indicated in the module.

The microscope

NATURAL SCIENCES

Grade 9

PLANTS

Module 25

THE MICROSCOPE

Plants, like all other living organisms, consist of basic elements, or cells.

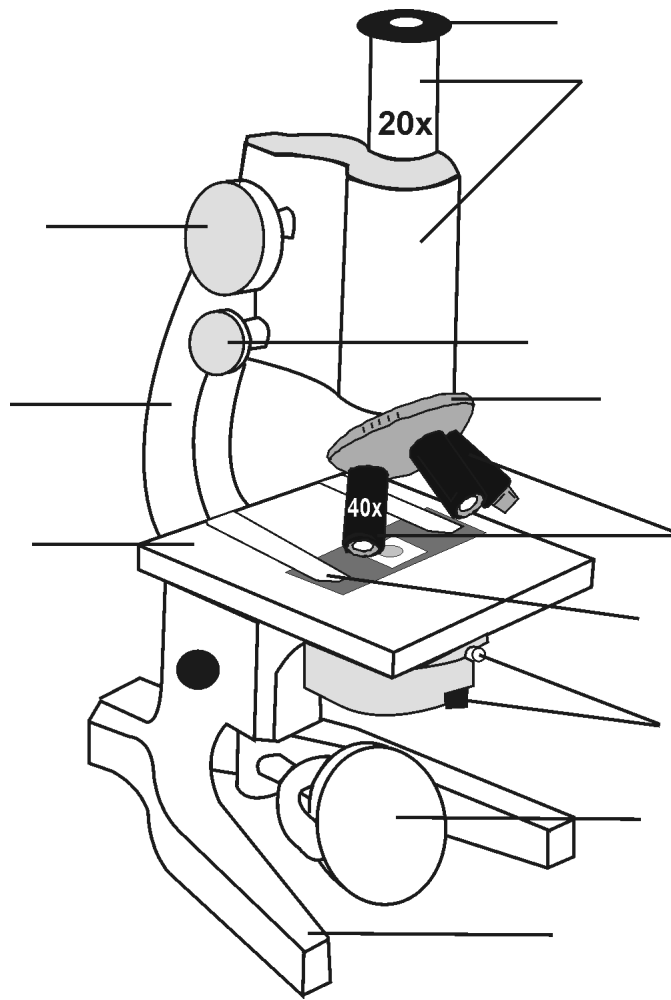
The structure of cells can be examined through a microscope. To be able to do this, it is necessary to learn how a microscope is operated. This will also unlock further possibilities in your studies.

Activity:

To explain the operation of a microscope

[LO 1.2, 1.3]

Your educator will introduce you to the parts of a microscope to show you how it is constructed and how it is used, and to give you the rules for when you use such an expensive piece of apparatus.



Activity:

To identify the different parts of a microscope

[LO 2.1, 2.2, 2.4]

Assignment 1:

1. Complete the labels for the sketch.
2. Write down the functions of the parts:

	Part of Microscope
1.	Eyepiece / ocular
2.	Barrel
3.	Arm
4.	Coarse adjustment screw
5.	Fine adjustment screw
6.	Rotating nose-piece / objective turret
7.	Objectives / lenses
8.	Clamps
9.	Stage
10.	Diaphragm
11.	Condenser
12.	Light bulb or mirror

3. Write down some of the important rules for handling a microscope:

4. Explain how to determine magnification:

5. What is the magnification to which the microscope that you are using has been set? Show your calculation.

Assessment: The functions of the microscope

Were you able to identify the names and functions of the basic parts?

[LO 1.2; LO 1.3]

Activity:

To use the microscope as apparatus

[LO 1.2, 1.3]

Assignment 1 Work together in groups to do the following:

1. Use a transparency pen to write on an object slide and examine the writing, using a low magnification.
2. Tear a small strip of paper, place it on the object slide and examine it through the microscope.
3. Select something of your own choice: a small dead insect, perhaps an ant, or one of your own hairs, for examination.
4. Examine a few grains of sugar.

Write notes on what you observe.

Assignment 2:

Prepare a wet specimen from a piece of the epidermis of an onion leaf. Your educator will demonstrate the technique for doing this. First use a low magnification for studying the specimen and then increase the magnification.

Draw the specimen as seen with the naked eye, and then as seen through the microscope.

The brick-like structures are **cells**. The whole body of any plant or animal (including people) is built up of **cells**.

Assignment 3:

You are now experienced in handling a microscope. Write down any suggestions that you would like to mention about handling the microscope and adjusting it for finding the correct image.

Why might you want to use a colorant to stain the specimen?

Ask your educator about a suitable colourant.

Assessment: The practical use of the microscope

Were you able to handle the items and complete the sketches?

[LO 1.2; LO 1.3]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- conducts investigations and collects data;
- evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;
- categorises information;

- interprets information;

2.4 applies knowledge.

Memorandum

MINIMUM CONTENT

Activity 1

Assignment 1:

- Plants, like all other living organisms, consist of basic elements or cells.
- The structure of cells can be examined through a microscope. To be able to do this, it is necessary to learn how a microscope is operated. This will also unlock further possibilities in your studies.

HOW A MICROSCOPE IS CONSTRUCTED

FUNCTION
Lens at the upper end of the barrel facilitates enlargement
Beams of light travel through this barrel, which separates the ocular and the objectives
Handle for holding
Moves the stage towards the object lens; Initial focussing
For finer, closer focussing

For moving objectives into position
These lenses make it possible to achieve different magnifications
Holds the microscope slide in position
For the positioning of the microscope slide
Controls the amount of light that passes from the object to the eye
Concentrates light on the object
Light source

How to handle a microscope

1. Carry it carefully – using both hands.
2. Take care not to mark it with fingerprints.
3. Never remove the lenses.
4. Handle the highest objective with great care.
5. Protect the microscope from dust and damp.

Magnification

- Value on the objective x value on the ocular
- E.g.: $10 \times 40 = 400 \times$ magnified

Activity 2:

Assignment 1:

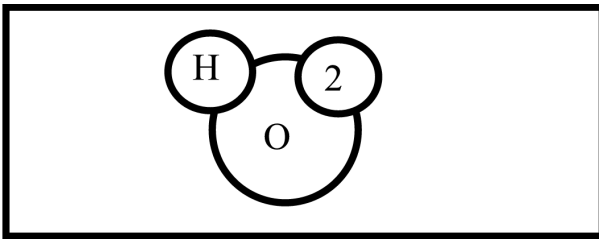
- Practical

Assignment 2:

Wet specimen

1. Select a microscope slide / object glass and wipe it clean.
2. Use a dropper to place a single drop of water in the centre of the slide.

Water drop Microscope slide



3. Using a tweezer or dissecting needle, select the object (specimen) and place it in the drop of water. Open it up and flatten it.
4. Stand a dissecting needle vertically in the centre of the object.
5. Position a cover-glass at an angle against the needle.
6. Quickly remove the needle to allow the cover-glass to fall into place over the specimen.
7. Tap the cover-glass lightly, using the back of the needle, to release bubbles that might be trapped.
8. Use an iodine solution to stain the specimen for greater cell definition. Starch grains will also be stained.

Structure of a plant cell

NATURAL SCIENCES

Grade 9

PLANTS: STRUCTURE, INCLUDING GENETICS

Module 26

STRUCTURE OF A PLANT CELL

Activity:

To understand the structure of a plant cell

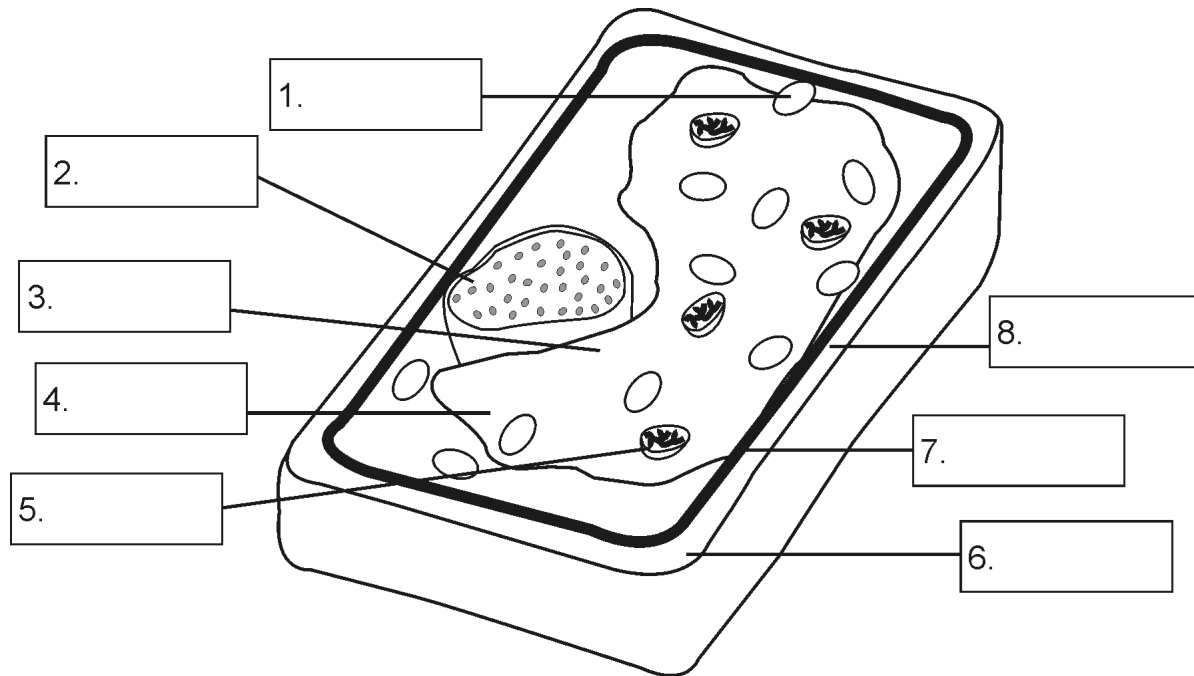
[LO 2.1, 2.2, 2.3, 2.4]

Ask your educator for a leaf from a moss plant, or some other example, to study through a microscope.

You could study the structure of a typical leaf. The following illustration is a simple line drawing providing a two-dimensional image of what you might see:

The cell consists of a framework filled with a special liquid – **PROTOPLASM** – with a variety of structures floating in it. Some of these are too small to be seen through an ordinary microscope. We need to use an electron microscope to examine these small structures. Remember that you see the cell from one side only. It is actually shaped like a brick and we say that it is three-dimensional. Ask your educator to explain this concept to you.

Find out about microscopes and electron microscopes.



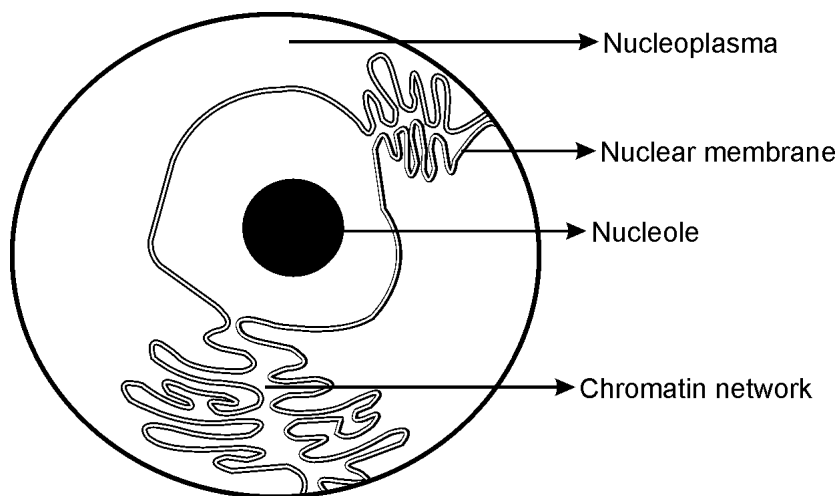
Assignment 1:

1. Provide a caption for the sketch.
2. Add labels in the numbered boxes.
3. What are the functions of each?

LabelNumber	Labels
1.	
2.	
3.	
4.	

5.	
6.	
7.	
8.	

Assignment 2: The nucleus contains fine details that you need to get to know before working through the rest of the module:



Discuss each of the labels of the nucleus in class and summarise their functions.

	Label
1.	Nucleole

2.	Chromatin network
3.	Nuclear membrane
4.	Nucleoplasm

Can you remember the main function of the nucleus?

Although the nucleus is the control centre of the cell, there are two other organelles that require closer investigation because of the important functions that they perform.

Plants engage in **PHOTOSYNTHESIS** and **RESPIRATION**. The organelles that are involved in this are the **CHLOROPLAST** (photosynthesis) and the **MITOCHONDRION** (respiration). You will be learning more about these organelles in future.

Your educator will show detailed sketches of these organelles to introduce them in greater detail.

Assessment: The identification of cellular structure

Were you able to distinguish the basic structures?

[LO 2.1; LO 2.3]

Assignment 3:

Make simple line drawings of the organelles,

Which of the structures and aspects that form part of the plant cell would you **not** expect to come across in an animal cell? Supply reasons:

(Suggestion: **Look** at yourself and at a plant, and **feel** what you are like and what a plant is like.)

Assessment: Line drawings and deductions

Were you able to do the sketches and make deductions? [LO 2.3; LO 2.4]

By now, you probably understand that plant cells have particular characteristics but that they do not all look the same, because they are adapted to fulfil specific functions. These adaptations in their structure are referred to as **DIFFERENTIATION** and the accompanying change of function is known as **SPECIALISATION**. Both terms are important in cellular studies.

A group of cells that are adapted to perform a particular function is called a particular kind of **TISSUE**. Some conduct water through the plant. Other groups of cells are responsible for strengthening the plant. More, still, are adapted for photosynthesis.

Tissues are grouped together to perform a combined function. Such a structure, consisting of many different kinds of plant tissue, but with a combined main function, is known as an **ORGAN**, e.g. a leaf.

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- conducts investigations and collects data;
- evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;

- categorises information;
- interprets information;

2.4 applies knowledge.

Memorandum

PLANT CELL

- It consists of a framework containing a special liquid – **PROTOPLASM** –with a variety of structures suspended in it.

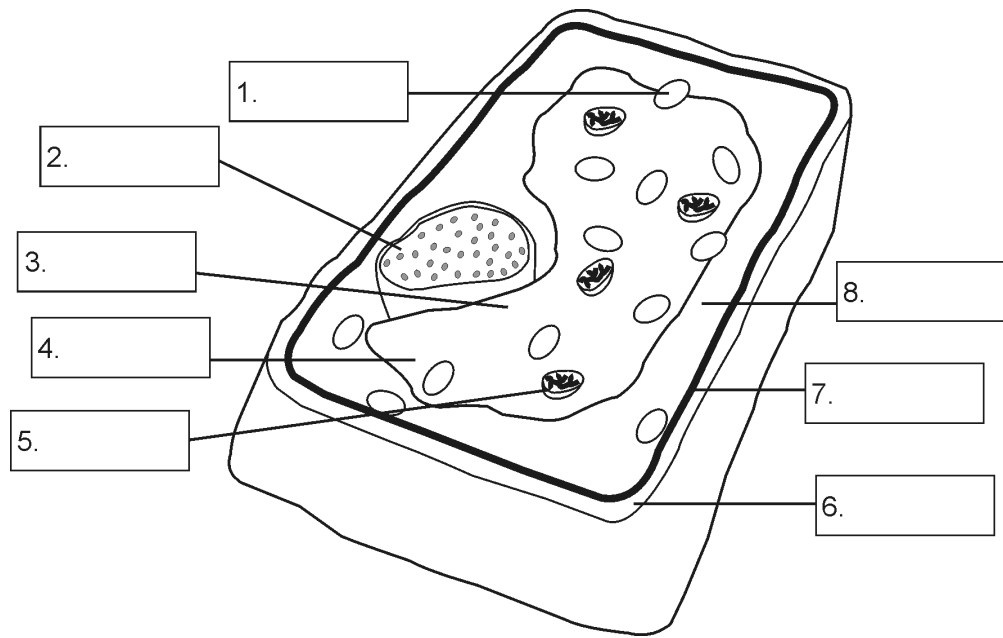
Assignment 3

- Control the light quantity carefully with the condensor/
- Focus carefully up and down when busy with the maximum enlargement.
- **Colouring:** with iodine solution for better self definition starch grains will also colour.

Activity

Assignment 1

1. Sketch of the plant cell:



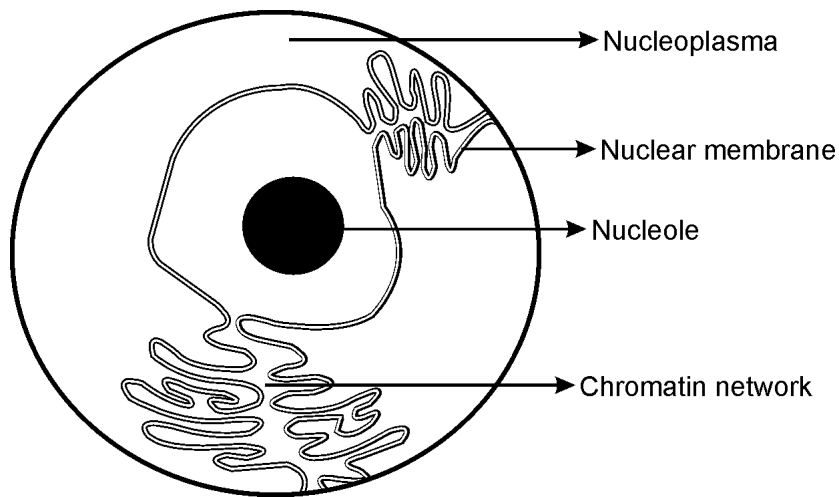
LabelNumber	Function
1.	Photosynthesis occurs here
2.	Controls all cell activity
3.	Membrane enclosing the vacuole; vacuole stores moisture and colourants
4.	Moisture, colourants
5.	Respiration, produces energy
6.	External protective and strengthening membrane
7.	Controls entry and exit of substances

8.

Medium in which reactions occur and in which the organelles are suspended

Assignment 2:

THE NUCLEUS



	Function
1.	Controls molecules in the nucleus
2.	Stores genetic material; bears hereditary characteristics
3.	Contains nuclear material, communicates with the rest of the cell
4.	Special structures and moisture

Assignment 3

- Plants **PHOTOSYNTHESISE** and **RESPIRE**. The two organelles that are involved are the **CHLOROPLAST** and the **MITOCHONDRION**.
- Studying this aspect of natural science remains important right through to matric.

CHLOROPLASTS

Contain the colour pigment **chlorophyll**;

Accommodate photosynthesis;

Enclosed by a double membrane;

Contains stroma of liquid and grana – stacked hollow disks (thylacoids) that are suspended in it. Grana are attached to other flat membranes (lamellae).

MITOCHONDRIA

accommodate the ENERGY-providing process of RESPIRATION;

also enclosed by a double membrane;

a fold/ridge of the inner membrane is called a crista (cristae);

the matrix is the liquid inside the mitochondrion.

- All the organelles that might occur are not discussed in Grade 9, but those that are mentioned will indicate some of the differences between typical plant and typical animal cells.

Absent in animal cells; maintains form and provides sturdiness in plant cells, as plants have no supporting skeletons.

Small or absent in the case of animal cells; it provides storage and also adds to the sturdiness of the plant cell.

Animal cells have no PLASTIDS.

Plant cells have a fixed shape determined by the cell wall – animal cells come in many shapes.

- Perhaps you have realised that plant cells are unique. In nature, though, there is endless variation in the shape and characteristics of plant cells. Such variation of form is referred to as **DIFFERENTIATION**, the variation in function that accompanies it is referred to as **SPECIALISATION**. Both of these terms are of importance in the biological study of cells.
- When cells are grouped together, we have to do with the next level of organisation, namely **TISSUES**. Tissues are cells that are grouped together to execute a specific task.
- This, for example, provides for cells that only serve to conduct water through the plant – the xylem.
- Tissues are also grouped together to perform a combined function. Such a structure, consisting of many different kinds of plant tissue, but with a combined main function, is known as an **ORGAN**.

Plant organs: roots

NATURAL SCIENCES

Grade 9

PLANTS: STRUCTURE, INCLUDING GENETICS

Module 27

PLANT ORGANS: ROOTS

Plants live and grow unobtrusively and we are often unaware of the marvels that are hidden within them. We need to take a closer look to understand how miraculously they function.

A plant consists basically of four kinds of organs: **roots**, **stems**, **leaves** and **flowers**.

Activity 1:

to identify the organs of a plant

[LO 2.2]

Assignment 1:

Complete the tables by supplying the missing information about the different organs.

Plant organ

Root
Stem
Leave
Flower

Above ground/Underground

Main function

Activity: 2

To determine how the structure of a plant stem is related to the functions that it performs

[LO 1.2, 2.4]

Plant roots

Assignment 1:

Go to the garden or the school playing fields and collect one small example of each of four kinds of common plants. Choose weeds, if possible. Ensure that you remove the plant with its root system intact. Rinse off any soil and arrange the plants on a sheet of clean paper.

Do you observe any noticeable differences with regard to the root system?

If you are lucky your plant samples will have two different kinds of root system. Sketch the two main types of root system. Your educator will provide the necessary guidance.

The **TWO** types of root system are known as the **ADVENTITIOUS ROOT SYSTEM** and the **TAP ROOT SYSTEM**.

Plants that have a taproot system have a main root that grows deep into the soil. Most of the plants that survive for more than a year (perennials) have a taproot system. This type of root system can reach water that is far below the soil surface and can anchor the plant. High trees and shrubs and bushes need such root systems. They belong to the group **DICOTYLEDONS**. They develop from seeds that have two seed leaves or cotyledons, like the seed of the bean plant. Many dicotyledonous plants, of course, are also annuals, but they do have taproot systems.

Monocotyledons, e.g. grasses and maize plants, are annuals. As they grow for one season only, they do not need very deep root systems. They actually need a wide, shallow root system to absorb all the moisture that might reach the soil. **MONOCOTYLEDONOUS** plants have adventitious root systems and develop from seeds that have one cotyledon, e.g. a maize kernel.

A **taproot system** consists of one main root, the **taproot**, and **lateral roots** that grow from it.

An **adventitious root system** consists of equally sized roots that grow from the base of the stem and are called **adventitious roots**. These roots are not branched.

Assignment 2:

Provide the correct labels for the sketches of the two types of root systems.

1. Which type of root system would you find with petunias, which are annuals?
2. Why do petunias develop this type of root system?
3. Which type of root system would an oak tree have?
4. Why would a tree need this type of root system?
5. Do trees have adventitious roots? Provide an explanation for this.
6. Can lateral roots develop from adventitious roots? Explain your answer.
7. Explain the main functions of roots.

Assessment of Root systems

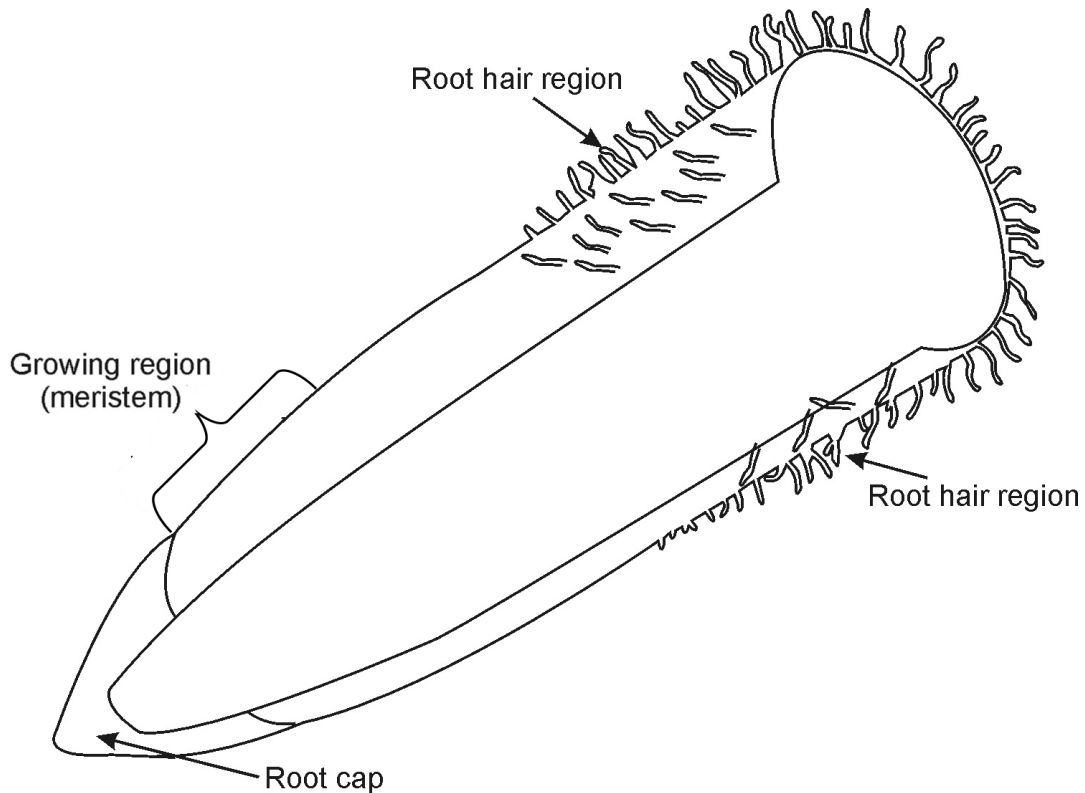
Were you able to distinguish different types of roots and apply your knowledge?

[LO 1.2, 2.4]

Take a fresh plant with roots, rinse the roots to remove the soil and carefully examine the roots. Are you able to see the fine, hair-like growths at the tips of the roots? They will be particularly visible on bean plants that you germinate from seeds yourself. These structures are known as root hairs, or radical hairs.

What is the function of a root hair?

The following is an illustration of the tip of a root:



Assignment 3:

Explain what happens in each of the following regions:

Root hair region:

Growing region:

Root tip:

Root hairs consist of special cells on the outer surface of the root right behind the growing region, which increase the surface of the root so that the maximum amount of water and mineral salts can be taken up from the soil.

Assignment 4:

Make a drawing showing the structure of a typical root hair. Provide your drawing with labels and a caption.

The root hair penetrates the small spaces between the particles of soil to suck up the moisture that is trapped between the soil particles. Then it is the function of the next plant organ, namely the plant stem, to conduct the moisture further.

Assessment: Root hairs

Were you able to draw the basic sketch?

[LO 1.2]

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- conducts investigations and collects data;

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- categorises information;

2.4 applies knowledge.

Memorandum

PLANT ORGANS: ROOTS

Activity 1

Assignment 1

- 1. Plants live and grow unobtrusively and we are often unaware of the marvels that are hidden within them. We have to take a closer look to learn to understand how miraculously they function.

A typical plant consists of the **PLANT ORGANS:**

Root, Stem, Leaf and Flower

Plant organ
Root
Stem
Leaf
Flower

Above ground or underground
Under
Above
Above
Above

Main function with regard to the plant	
Absorbs water and minerals; anchors the plant.	
Bears the leaves and flowers, and fruit, eventually, in a way that exposes each organ correctly.	
Photosynthesis	
Propagation	
Above ground or undergroundMain function with regard to the plant	
Root UnderAbsorbs water and minerals; anchors the plant.	
Stem AboveBears the leaves and flowers, and fruit, eventually, in a way that exposes each organ correctly.	
Leaf AbovePhotosynthesis	
Flower AbovePropagation	

Activity 2

Assignment 1

ROOT SYSTEMS

1. There are adventitious root systems and taproot systems.
2. Plants develop one or the other of these systems according to the type of plant that they are (There are two main types).
3. Plants that survive for more than one year, perennials, require deep root systems for finding water during dry seasons and to be anchored properly. You will realise that tall trees and large shrubs and bushes need such root systems. They belong to the group called **DICOTYLEDONS**. They develop from seeds that have two seed leaves or cotyledons, like the seed of the bean plant.
4. Other plants, like grasses and maize plants, annuals, live and grow for one season only and therefore do not need such deep root systems. They need wide-reaching, shallow root systems. These plants belong to the **MONOCOTYLEDONS** and develop from seeds that have one seed leaf, e.g. a maize kernel.

Assignment 2

A **taproot system** consists of one main root, the **taproot**, and **lateral roots** that grow from it.

An **adventitious root system** consists of equally sized roots that grow from the base of the stem, the **adventitious roots**.

The life cycle of annuals is completed in a year – they do not need taproots – and they will therefore have an adventitious root system

Trees have a taproot system because they require to be anchored deeply and have to obtain water from deep below the ground.

As adventitious roots grow from stems, they will be of no value for trees that need deep anchoring.

Lateral roots only grow out of other roots.

Main functions of roots:

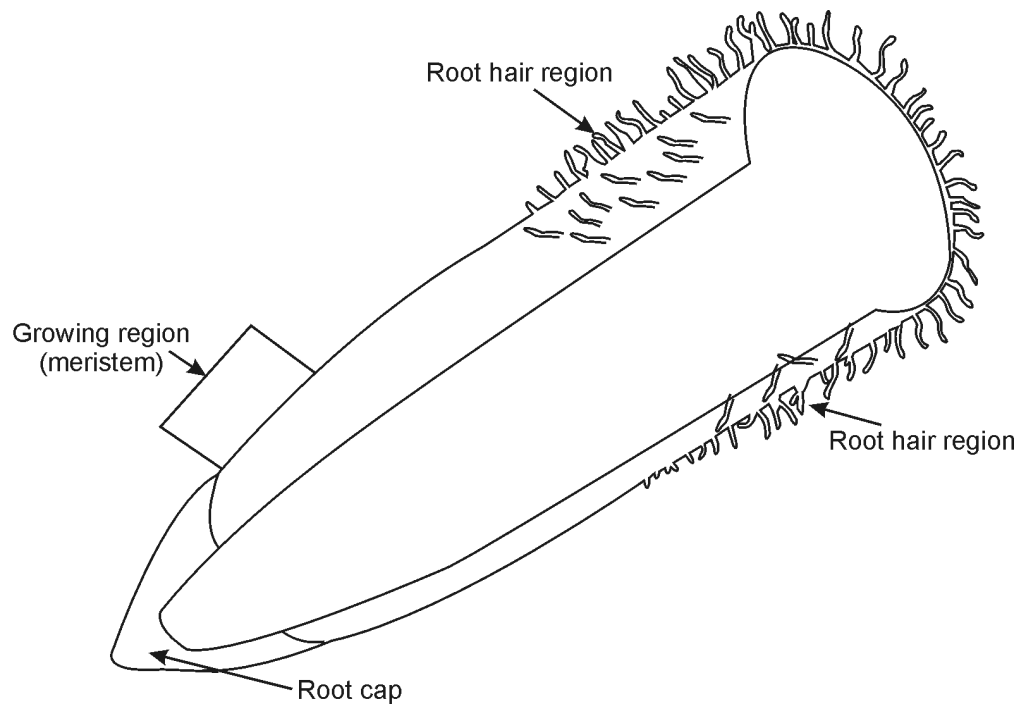
anchoring

uptake of water

uptake of mineral salts in solution

conducting water to the upper parts of the plant

When a root system is investigated, the following structure can be observed:



Assignment 3

Root hair region: region where root hairs develop

Cell lengthening region: region where cells lengthen and differentiate

Cell division region: where cells divide

Meristem: general region where cell division and cell lengthening occur

Root tip: Protects the sensitive foremost growth tip by keeping it slimy to prevent wear.

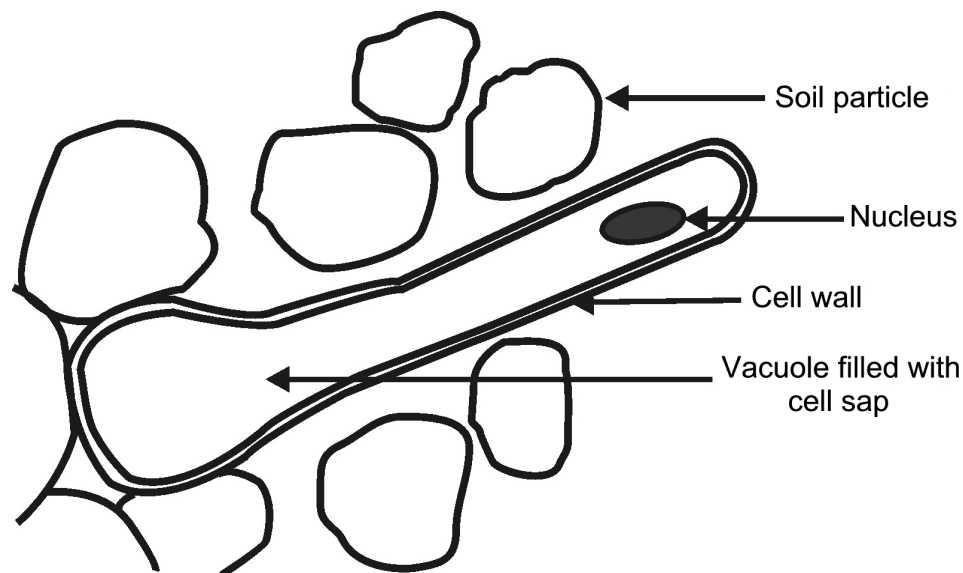
1. A root hair is a special cell on the outer surface of the root – immediately following the growth tip – that is specially enlarged to take up a maximal amount of water and salts from the soil.

The structure of the root hair is of importance in further studies in Biology.

Assignment 4

This is what we shall focus on next.

Root hair



1. The root hair penetrates the small spaces between the particles of soil to suck up all moisture that is trapped between the soil particles. Then it is the function of the next plant organ, namely the plant stem, to conduct the water.

Plant organs: stems

NATURAL SCIENCE

Grade 9

PLANTS

Module 27

PLANT ORGANS: STEMS

Activity:

To determine how the structure of a plant stem is related to the functions that it performs

[LO 2.1, 2.2, 2.3, 2.4]

The functions of the plant stem include bearing the leaves, flowers and fruit in such a way that they are exposed to sunlight, pollinating agents and fruit distributors.

In the case of large trees, the stem, in time, becomes the trunk, which is greatly thickened and strengthened. The organs that are above ground cannot live without water. As water is obtained from the soil, it has to be conducted upwards. The conducting channels for water and mineral salts are formed by lengthened, dead cells, which form the wood of the plant. The botanical name for wood is **xylem**. The nutrients have to be conducted to the rest of the plant, including the roots. A system of living, lengthened cells forms the bark of the tree. The botanical name for bark is **phloem**.

Leaves, in their turn, manufacture starch and sugar.

Assignment 1:

Collect a few twigs and stems from different plants. Place these on the desk and examine them for differences.

Divide your specimens into two groups: brown, woody types and green herbaceous types.

1. Why do these differ?

This depends on the type of plant.

1. What kind of stem would an annual, monocotyledonous plant have?
1. What kind of stem would a perennial dicotyledon, e.g. a fruit tree, have?

Fruit trees are deciduous.

1. What does this mean?
1. Leaves make it possible for a plant to "breathe". What happens in the case of deciduous trees that lose their leaves during the winter? How do they "breathe"? (They do not really breathe, of course. We should ask how gas exchange takes place.)

Take a look at the brown, woody twigs that you have collected. There is a fair chance that you will notice white specks on the brown twigs. These are **LENTICELES**. Lenticels are pores (openings) that facilitate gas exchange during winter.

Assessment: Study of stems

Were you able to recall information that enabled you to distinguish what you needed to know?

[LO 2.1]

Assignment 2:

1. Draw an example of two different types of twig – a green, herbaceous type and a brown, woody type.
2. Your educator will assist you with the descriptions of the following concepts. They are related to structures found on stems.

Terminal bud:

Apical meristem:

Internodes:

Nodes:

Leaf nodes:

Axil:

Scale leaves:

3. Refer to the above list (at no. 2) to provide labels for the illustrations in no. 1.

4. Make your own deduction: What are the main functions of stems?

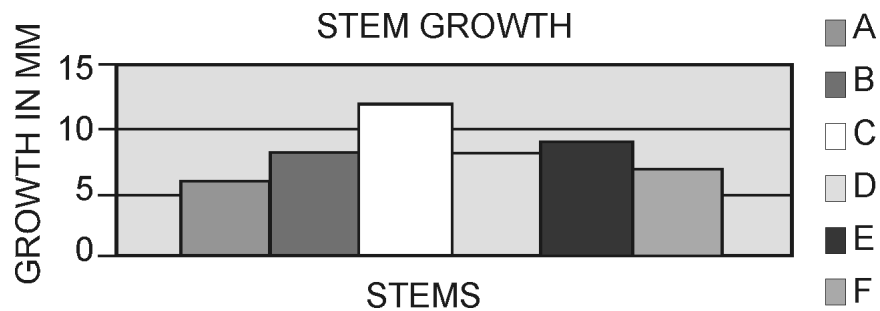
Assessment: Sketches and Questions

Were you able to indicate the differences and the structure?

[LO 2.2; LO 2.3]

Assignment 3:

The following column graph shows the longitudinal growth of six different sunflower seedlings (A, B, C, D, E, F) during the first week after germination.



1. What is the difference between a column graph and a histogram?
1. Which of the seedlings experienced the most favourable conditions for growth? Provide a reason for your answer.
1. What height did seedling A attain in the first week?
1. Which two seedlings displayed equal longitudinal growth?
1. What was the difference in longitudinal growth between seedlings B and F?
1. What was the difference between the longest and the shortest seedling?
1. Describe the way in which a seedling's longitudinal growth occurs.

Assessment: Graph

Were you able to answer the questions?

[LO 2.3]

Assessment

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;
- categorises information;
- interprets information;

2.4 applies knowledge.

Memorandum

Plant organs: Stems

Activity

Assignment 1

- As you have indicated, the functions of the plant stem include bearing the leaves, flowers and eventually also the fruit, in such a way that they are exposed to sunlight, pollinating agents and fruit distributors respectively.
- In the case of large trees, the stem, in time, becomes the trunk, which is greatly thickened and strengthened with bark and other structures.
- Plant stems need to have channels for conducting the main products that have to be moved up and down. If you understand that water has to be conducted from the lowest tip of the roots to the highest tip of the uppermost leaf, you will realise that we need a conducting channel for such water and for mineral salts.
- The leaves produce starch and sugar, which have to be conducted to all the cells, even to those that are at the furthest tips of the roots, but cannot be conducted along the same channels as the water, because this would lead to confusion. The special channel for these nutrients is known as the **PHLOEM**.
- Further variations occur on the outer surfaces of stems.
- Herbaceous stems are green and fleshy, but woody stems are brown and hard.
- The annual monocotyledon usually has fleshy, green herbaceous stems and the perennial dicotyledon, e.g. a fruit tree, has hard, woody stems.

- As we are mentioning fruit trees – they are deciduous, which means that they lose their leaves in autumn.
- If you know that leaves are the most important part of the plant (and therefore of fruit trees) when it comes to respiration, and that deciduous trees are without leaves during winter, you might ask what they do about respiration. We should actually ask how gas exchange takes place.
- Examine the brown, woody twigs that you have collected. There is a good chance that you will notice fine white specks on these brown twigs. These are **LENTICELS**. Lenticels are pores (openings) that facilitate gas exchange during the wintry season.

Assignment 2

DIFFERENT PARTS ON THE TIP OF STEM

Terminal bud: end bud of a stem, where the growing tip is situated.

Apical meristem: growth tip tissue at the end tip.

Internodes: regions of the stem between nodes.

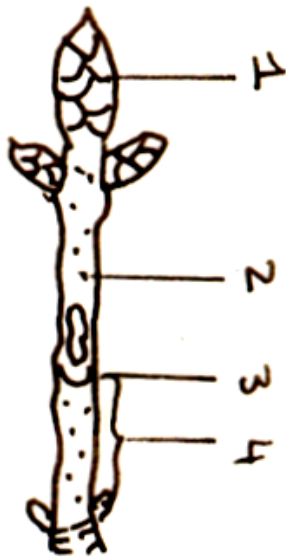
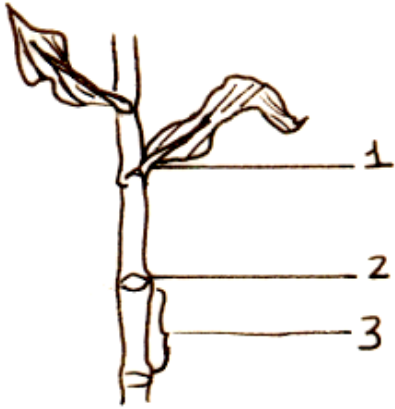
Nodes: region on stem where buds, leaves and lateral shoots will develop.

Leaf nodes: marks on stems where previous leaves have fallen off.

Axil: region in a node where bud growth will take place.

Scale leaves: Protective leaves for the terminal bud and other growth tips.

1. different types of twig – a green, herbaceous type and a brown, woody type.



Main functions of stems:

to carry leaves in such a way that photosynthesis can take place;

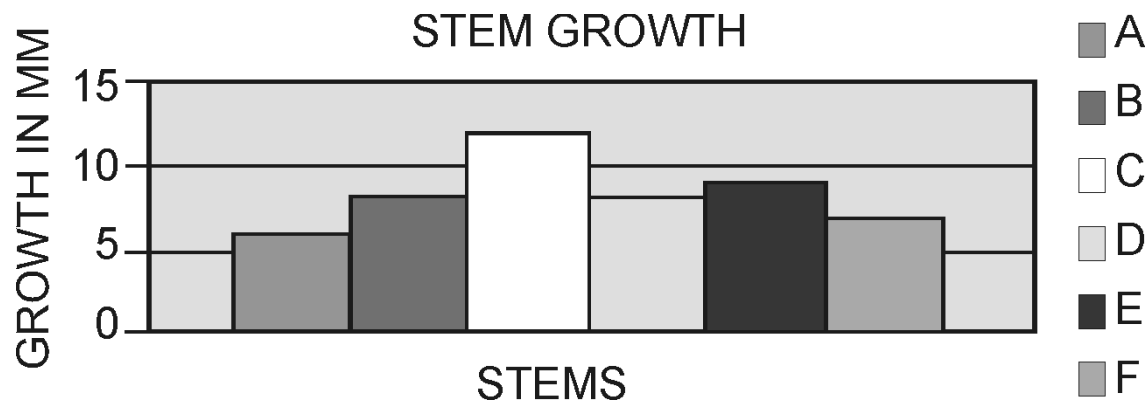
to carry flowers in such a way that they can be pollinated;

to carry fruit in such a way that they can be dispersed;

to conduct water and nutrients between leaves and the roots.

Assignment 3:

- The following column graph shows the longitudinal growth of six different sunflower seedlings (A, B, C, D, E, F) during the first week after germination.



1. histogram have spaces inbetween
2. M – highest growth
3. approximately 6 mm
4. L and N
5. $8 - 7 = 1\text{mm}$
6. $12 - 6 = 6\text{mm}$
7. cells divide – cells lengthen and differentiate = growth

Plant organs: flowers

NATURAL SCIENCES

Grade 9

PLANTS

Module 29

PLANT ORGANS: FLOWERS

Flowering plants (Angiosperms) are the most prolific and successful group of plants on earth. Their secret is partly related to the way in which they are propagated. The key to this secret lies in the structure of the flowers and in how they operate. The fact that propagation is **SEXUAL**, adds to their success.

The flower is a structure that contains both the **male and female structures** of the plant. Male and female structures do sometimes occur in separate plants, but they mostly occur in a single **bisexual (gynandrous)** flower.

Activity 1:

To describe flowers as structures for propagation

[lo 2.2, 2.3]

Assignment 1:

1. Take a flower and make a lengthwise cut through it to divide it in two. The *Petunia* is a good example to use. Make a drawing of the flower, showing all its parts (without labels).

Name the functions of each of the following (find the answers yourself or enlist the educator's help):

Anther:	
Filament:	
Stigma:	
Style:	
Ovary:	
Seedbud / ovulum:	
Petal:	
Sepal / calyx-leaf:	
Stamen:	
Pistil:	
Footstalk / peduncle:	

1. Try to identify all the parts listed in no. 2, in your sketch for no. 1.

4. Answer the following questions:

What is the term we use to refer to the group of plants that bear flowers?

Are all types of flowers equally colourful? Provide an explanation.

Are all flowers fragrant? Provide an explanation.

What is the function of all flowers?

Assessment: Sketches

Were you able to distinguish the basic differences?

[LO 2.2; LO 2.3]

Activity 2:

To describe pollination as a necessary process for the propagation of flowering plants

[LO 1.1, 1.2, 1.3, 2.4]

Pollination is the process by which a mature (ripe) grain of pollen lands on the mature pistil of a similar plant.

Assignment 1:

1. Why must the grain of pollen be mature?
2. Why should the pistil be mature and receptive?
3. Why must the plant that is pollinated be similar to the one from which the pollen is derived?

The pollen of each type of plant has a unique structure and are microscopically different from the pollen of the plants. This means that a grain of pollen has to fit like onto the pistil like a key in a lock.

A ripe grain of pollen will begin to germinate as soon as it has landed on an appropriate surface.

Assignment 2:

Undertake an investigation to obtain information about the pollen and pistils of different kinds of plants.

Assemble a sheet of sketches of the different kinds of pollen and pistils that you discover.

Assessment: Investigation

Were you able to distinguish basic differences?

[LO 1.1; 1.2; 1.3]

Assignment 2:

Design a flower of your own with the help of the following guidelines:

Your flower is of a very rare, exotic type. It grows in the upper reaches of mountains where there are hardly any insects. It has to rely on wind for pollination. The mechanism by which pollination occurs will be the absolute deciding factor for what the flower's structure will be like.

Assessment: Flower design

Were you able to distinguish basic differences?

[LO 2.4]

Activity 3:

To describe fertilisation as a necessary process for the propagation of flowering plants

[LO 2.1, 2.2]

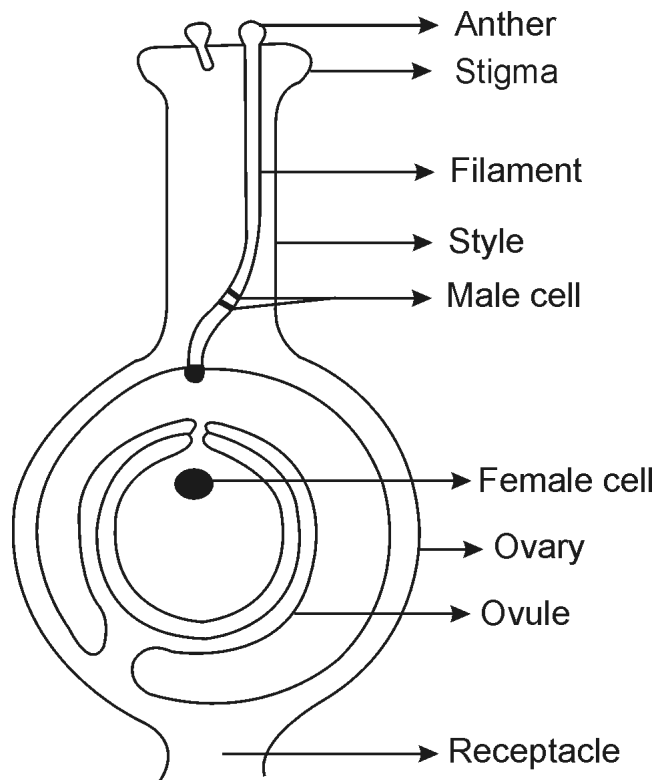
We have mentioned that two main processes during reproduction occur, namely pollination and fertilisation. We know what pollination is, and that the ultimate goal of the grain of pollen is to reach the mature ovule to fertilise it.

Fertilisation is the union of the special nuclear contents (gametes) of a grain of pollen and an ovule.

The nuclear content comprises a special molecule that has a code or “plan” of what the new plant must be like. We refer to this as the chromatin network, which consists of the molecule known as DNA. You will learn about the exact significance of this later.

A zygote, the beginning of the new organism, or plant, develops as soon as the nuclear content or gametes have fused.

The process of fertilisation is represented in the following sketch:



Assignment 1:

1. Present the five-step process that takes place from the moment of pollination to fertilisation:
2. Explain what happens to the following structures once fertilisation occurs:
 - a) ovary
 - b) ovule

Assessment: Mechanism of pollination and fertilisation

Were you able to analyse the mechanism and present its steps?

[LO 2.1; LO 2.2]

The formation of seeds is necessary for the process of propagation. The process of distributing the seeds or fruit ranks with this in importance.

Optional / For Enrichment:

The educator could suggest a project dealing with fertilisation through which you could investigate the interesting ways and mechanisms of propagation that form part of this process.

The David Attenborough video, *The Secret life of Plants*, could be viewed as an alternative. It contains astonishing visual material.

Assessment

LO 1: Scientific investigations:

The learner will be able to act confidently on curiosity about natural phenomena, and to investigate relationships and solve problems in scientific, technological and environmental contexts.

This is evident when the learner:

- plans investigations;
- conducts investigations and collects data;
- evaluates data and communicates findings.

LO 2: Constructing Science Knowledge:

The learner will know and be able to interpret and apply scientific, technological and environmental knowledge.

This is evident when the learner:

- recalls meaningful information;
- categorises information;
- interprets information;

2.4 applies knowledge.

Memorandum

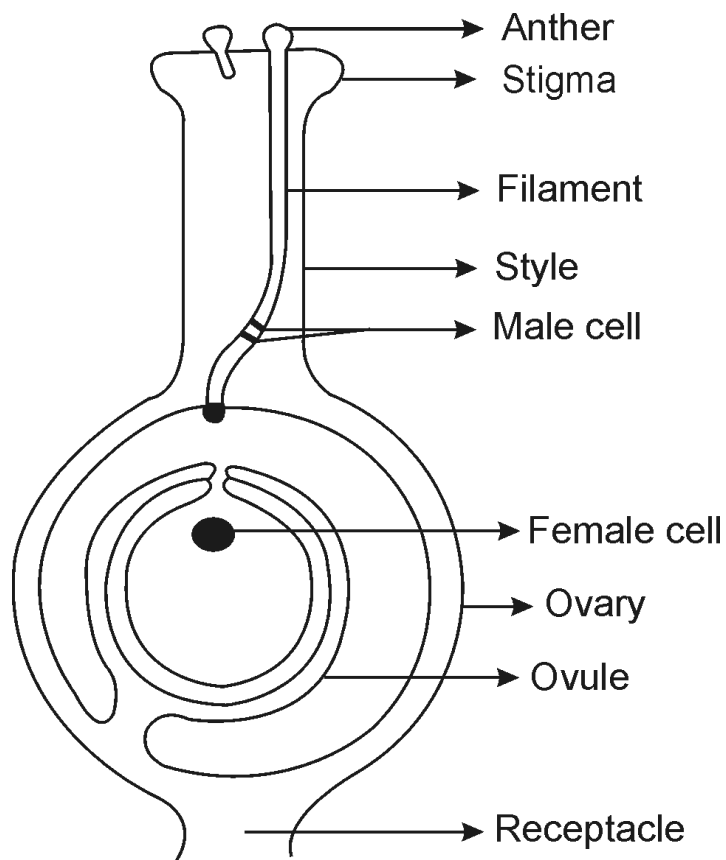
PLANT ORGANS: FLOWERS

Activity 1

Assignment 1

- Flowering plants (Angiosperms) are the most prolific and successful group of plants on the earth. Their secret is partly related to the way in which they are propagated. The key to this secret lies in the structure of the flowers and in how they operate. The fact that propagation is **SEXUAL**, facilitates their success.
- The flower is a structure that contains both the **male and female structures** of the plant. It is true that these aspects sometimes occur in separate flowers, but they generally occur together in a single **bisexual (gynandrous)** flower. If not, they will be unisexual, which means that there will be male and female flowers.

1. Make a drawing of the flower, showing all its parts



2. Name the functions of each of the following:

Anther:	where pollen is produced
Filament:	the stalk of the stamen, which holds the anther in the right position
Stigma:	the part of the ovary where the ripened pollen has to fall
Style:	the long extension of the ovary that bears the stigma
Ovary:	the place where the ovules are formed

Seedbed / ovulum:	the body that develops into a seed when it is fertilised
Petal:	part of the corolla, which attracts insects (by colour)
Sepal / calyx-leaf:	protects the flower in its bud stage
Stamen:	the male reproductive part of a flower, consisting of the anther and the filament
Pistil:	the female reproductive part of a flower, consisting of the stigma, style and carpel
Footstalk / peduncle:	the small stalk that bears the flower in the right position.

3. Augiosperms

4. No, only those that are pollinated by insects

5. No, only those that have to attract insects

6. Propagation of the specis – fertilisation of the ovaries

- Angiosperms or flowering plants, especially those with brightly coloured petals, attract insects and are pollinated by them. They are also fragrant.
- The function of the flower is to ensure the fertilisation of the ovules.
- The main functions therefore are **POLLINATION** and **FERTILISATION**.

Activity

Assignment 1

POLLINATION

Pollination is the process by which a ripened (mature) grain of pollen is deposited on a mature stigma of the same type of plant.

- A grain of pollen must be ripe, or it will not be able to germinate to grow into the style. The stigma also has to be ripe and receptive to the pollen to be able to develop. The pollen has to come from the same sort of plant, or crossbreeding will result. Sometimes crossbreeding is possible, and this can result in all kinds of exotic plants.
- The structure of the grains of pollen of each type of plant differs microscopically from those of all other kinds of plants. This means that a grain of pollen virtually has to fit like a key in a lock. A ripe grain of pollen will begin to germinate as soon as it has landed on an appropriate surface.
- We have mentioned the fact that two main processes have to occur, namely pollination and fertilisation. We know what pollination is and that the ultimate goal of the grain of pollen is to reach the mature ovule to fertilise it.

Assignment 2:

- Undertake an investigation to obtain information about the pollen and pistils of different kinds of plants.
- Assemble a sheet of sketches of the different kinds of pollen and pistils that you discover.

Fertilisation is the union of the special nuclear contents (gametes or germ cells) of a grain of pollen and an ovule.

- The nuclear content that is referred to comprises a special molecule that has a code or “plan” of what the new plant must be like. We refer to this as the chromatin network, which consists of the molecule known as DNA. You will learn about the exact significance of this later.
- A zygote, the beginning of the new organism or plant, develops as soon as the nuclear contents or gametes have fused.
- The process, as it occurs step by step from the moment of pollination to fertilisation, comprises the following:

Assignment 3

- Design a flower of your own with the help of the following guidelines:

Own memo

Activity 3

Assignment 1

A ripe grain of pollen is deposited on a receptive stigma

Germination : growing down the a pollen tube into the style

Entering the carpel through the micropyle

Travelling through the carpel to the ovule

Two male gametes in the pollen tube are released

One fuses with the ovule's female gamete

FERTILISATION has occurred – a zygote is formed

The zygote develops into an embryo inside the seed

The carpel becomes the pericarp for seed distribution

Carpel: develops into a fruit

Seed bud: develops into a seed
Pistil: dries up and decays – see this at the bottom of an apple, for instance
Zygote: develops into an embryo

- Seed formation naturally is of the greatest importance to the process of propagation, but the process of distributing the seeds, or fruit, ranks with this in importance.

OPTIONAL / FOR ENRICHMENT

- The educator could suggest a project dealing with fertilisation through which you could investigate the interesting ways and mechanisms of propagation that form part of this process.
- The David Attenborough video dealing with “The Secret Life of Plants” could be viewed as an alternative. It contains astonishing visual material.